Brain2Body: The Next Generation In Body-Brain Interfacing.

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Abstract

An assisted hand can perform a movement without its nervous system having learned to perform it. Brain2Body asks what remains when teaching commands end, with any compensatory movement bridge held constant. The proposal connects stated nervous-system and body observations to independently measured outcomes, then separates assistance, simulated control, retained learning and biological repair. Published motor-control research motivates the experiments; the companion computational studies provide bounded comparisons rather than biological teaching evidence. In the current LINK study, 18 runs and 720 saved epochs fail both model-admission gates and all five benefit claims. The proposed next experiment pairs EEG, EMG and calibrated movement with a musculoskeletal physics engine and a learning model. Forward movement prediction and reverse recorded-feature reconstruction are separate tests; neither identifies a unique brain state. Sensory delivery requires a separately qualified writer. The comparison must separate what the model predicts, what the controller executes and what the trainee retains. This article develops the rationale, controls and limitations of that test. It reports no new biological experiment or retained skill acquisition.

1 Assistance and Retained Learning

Let a computer guide a hand through a tune, then stop the teaching commands. What can the trainee play now? The first performance shows what the combined system could do; the second begins to test what the nervous system retained. Any movement bridge needed to express the trainee's own control must remain available in both comparisons. Otherwise we would be testing a disconnected hand and calling the result a test of memory. Brain2Body connects nervous-system measurements to physical outcomes to make this distinction testable. The numeral 2 means "to"; body-to-brain feedback matters as much as the outward command.

Rapid acquisition of difficult physical skills is the ambition. Sewing, playing an instrument, manipulating a puzzle and coordinating a whole body put that ambition under pressure: each demands more than one correct movement. The programme must explain how useful guidance could become a capability the trainee retains. These are proposed tests, not demonstrated abilities. Neither arbitrary skill upload nor complete nervous-system reproduction is established, and the current models have not passed their original performance gates.

A decoder turns an observation into a prediction. Before calling it correct, we must decide what answer it is meant to give. NeuroTranslation proposes relating neural signals and their representations between explicitly described sources and targets;

Brain2Body follows that prediction into a body, where movement changes the feedback available for the next action and learning becomes a further question. The argument therefore needs an independently measured action, a model whose contribution can be isolated, and a retention test that can fail. Published research motivates those requirements. The companion computational work supplies bounded comparisons, including negative results; it has not supplied evidence of biological teaching.

2 Compare Predicted and Measured Movement

Perception is subjective. Movement is not. A person experiences a sensation; an independent instrument can measure a finger's position, a contact, a force or the timing of a release. That contrast does not make every movement measurement exact or reveal private experience. It gives the programme a tractable starting point: relate an uncertain neural observation to an outcome that can be checked.

Consider a key press. Choosing the right key, reaching it, pressing at the right time, releasing it and avoiding neighbouring keys are separate outcomes. A video can show position; a contact sensor can show a strike; a force sensor can measure force. A task score records success under declared rules. Neural activity near an electrode is another observation. None silently supplies the others.

In the proposed loop, sensory input guides action. The body and environment change; measured outcomes supply feedback for the next action. Feedback control adjusts action using the difference between the desired and observed outcome. Todorov and Jordan provide an established account of motor coordination under uncertainty. Applying it here is a programme rationale, not an empirical result of their paper. Every claimed loop must disclose its delays and whether its feedback came from a living subject, a simulation or a recorded replay. Todorov and Jordan, 2002.

Accuracy means agreement with an independent reference. Precision means repeatability under stated conditions. A consistently wrong model can be precise. Ground truth means an independently acquired measured reference with its own calibration, sampling limits and uncertainty; it is not error-free access to biology. Finer timesteps, more decimal places and a confident animation cannot establish these properties. NIST measurement terminology.

3 Structure, Activity and Function

A movement score tells us what happened. Explaining why requires more than a map of the routes a signal could take. Timing, the cells' responses to chemical signals, the body's current state, learning history and task conditions affect what follows. An intervention deliberately changes something to test its effect; a connection map alone cannot tell us that response. Structure narrows the explanation, while measured function must test it. That is why a body-grounded prediction needs both.

The Human Genome Project offers a bounded analogy. Obtaining a reference sequence did not finish the investigation of regulation and function; subsequent

projects investigated functional elements. That is a distinction between a map and its interpretation, not a claim that genome sequencing failed. NHGRI, Beyond Genes.

The MICrONS Consortium combined recorded visual responses with detailed connectivity in a volume of mouse visual cortex. Its strength is the combination of different measurements. It does not provide a whole human nervous system, peripheral or spinal signals, or an unrestricted account of learned skills. Co-registration means relating measurements to corresponding locations; such correspondence must be demonstrated for the actual source, not borrowed from an attractive template. MICrONS Consortium, 2025.

Measurement itself needs explanation. Functional magnetic resonance imaging, or fMRI, measures an indirect blood-related signal over time. In the experiments of Logothetis and colleagues, that signal related more closely to local field potentials, aggregate electrical activity near a recording site, than to spiking output. It is not a photograph of thought. Tang and colleagues demonstrated constrained semantic reconstruction with substantial dependence on the subject and cooperation; this is not universal one-to-one decoding of subjective experience. Logothetis et al., 2001, Tang et al., 2023.

Quantum measurement can illustrate that observing a system may involve interaction. It does not establish quantum backaction, the influence of a quantum measurement on a system, as the cause of these large-scale neural limits. Scanner constraints, indirect signals, person-specific learning and ambiguous representations supply concrete explanations that must be tested on their own terms. The analogy stops before it becomes an unsupported brain mechanism. Royal Swedish Academy, quantum measurement background.

4 Existing Brain–Body Interfaces

Bouton and colleagues linked recordings from motor cortex, movement-related tissue in the brain's outer layer, to electrical activation of forearm muscles, enabling useful hand and wrist movements in one participant with quadriplegia, paralysis affecting four limbs. This demonstrates a participant-specific route from brain signals to bodily action. It does not demonstrate an unfamiliar skill transplant or a complete map of the nervous system. Bouton et al., 2016.

Lorach and colleagues linked cortical recording to spinal stimulation and evaluated standing and walking in a participant with chronic spinal injury. They also reported rehabilitation-associated improvement with the interface switched off. Movement in a real participant provides functional evidence beyond a rendered demonstration. The study remains a particular participant, intervention and rehabilitation history; assistance, retained improvement and tissue regeneration are distinct claims. Lorach et al., 2023.

Flesher and colleagues used stimulation of sensory brain tissue to provide touch-related feedback during robotic arm control. The work supports testing the return path of an interface. It does not establish transfer of an expert's repertoire. These studies motivate bounded experiments; they cannot be stacked into proof of a

new integrated system whose relevant links have never been measured together. Flesher et al., 2021.

5 Digital-Fly Models

Digital flies sharpen the question of attribution: when a composite system moves well, which component supplied the ability? The answer determines whether the result tests neural wiring, a trained controller, body mechanics or their engineered combination. Those contributions must remain visible before a movement is credited to a reproduced nervous system.

FlyWire reconstructed 139,255 neurons, nerve cells, in one adult female fly brain. This is an exceptional structural resource with reconstruction and coverage limits. A map of chemical connections does not supply every connection's physiological effect, the complete nervous system, the body or the environment. Pospisil and colleagues propose combining intervention measurements with structural constraints to estimate causal effects; their estimator tests in simulation are not a completed empirical map of all fly-brain dynamics. Dorkenwald et al., 2024, Pospisil et al., 2024.

Shiu and colleagues show the positive route: specific connectome-derived predictions were tested through neural stimulation and behaviour. A connectome is a map of neural connections; it supplied constraints, not an automatic functional certificate. Useful predictions coexist with missed experimentally observed Phantom and Usnea effects and incorrect G2N-1/Usnea water-silencing predictions. The authors report 91% agreement across 164 tested predictions, or 84% after excluding one mostly negative screen. Neither figure measures whole-fly fidelity. Missing receptor dynamics, electrical connections and chemical modulation limit the account, including absolute simulated firing rates. Shiu et al., 2024.

Body mechanics introduce another layer. NeuroMechFly v2 documents simulated leg tips slightly below the ground because of numerical errors or physics instability, with a 0.05 mm margin to prevent false overstretch detection. Its adhesion model uses engineered forces and stepping phases while biological liftoff mechanisms remained uncertain. These are specific disclosed modeling limitations, not evidence of dramatic tunnelling or an ability no living fly could ever acquire. FlyBody uses trained artificial controllers to imitate measured movement; its time-averaged aerodynamic approximation and omitted detailed muscle and sensory physiology remain separate limits. Wang-Chen et al., 2024, Vaxenburg et al., 2025.

Eon's March 2026 technical account describes hand-chosen brain–body mappings, substantial reliance on existing body controllers, largely missing state and learning mechanisms, and internal dynamics not yet validated against relevant biological signatures. That dated developer disclosure supports a conclusion about an engineered integration, not a complete biological upload. We have not independently reproduced its dynamics. Eon Systems, 10 March 2026.

The same attribution problem appears in games. Pinned fly-chess code supplies symbolic board information, selected mathematical dynamics, a trained readout and external legal-move search. Pinned DoomFly code uses the distinct male central-nervous-system map and injects game-engine enemy geometry into model

neurons after the intended visual route reportedly failed. These observations come from inspected code and author accounts, not reproduced gameplay or evidence that a living fly learned either game. Fly-chess trainer, DoomFly threat input.

6 Simulation Failures

Attribution also depends on the world in which the controller succeeds. A controller can improve against a score while moving farther from the intended task. To use simulated practice as evidence, we need to know both what the score rewards and whether the simulated movement obeys the relevant physical constraints.

An optimizer, the procedure adjusting a model to improve its objective, can exploit an incorrect simulator or a poorly chosen score. First-hand historical accounts describe virtual creatures gaining propulsion from numerical errors and soft robots receiving artificial energy from ground-penetration corrections. CoastRunners illustrates a different failure: circling reward targets improved the score while leaving the race unfinished. Physics violation and objective misspecification need different diagnoses. Neither example establishes that every fly model shares those failures. Lehman et al., first-hand accounts, Google DeepMind, specification gaming.

Protein prediction offers another carefully bounded comparison. AlphaFold3's original paper describes invented structure in unstructured regions and overlapping predicted chains. Its antibody-interface benchmark reports 62.9% cluster-averaged success at DockQ > 0.23, a score threshold for agreement with experimentally determined interfaces, hence 37.1% failing that threshold. It covers 65 interface clusters from 71 complexes, with confidence-based selection from 1,000 seeds and five samples per seed for each target. This is not a universal percentage of proteins that fail to function. Crystallization is one measurement bottleneck; other methods include cryo-electron microscopy, which images samples using electrons, and nuclear magnetic resonance, which uses atomic responses to magnetic conditions. Synthetic examples can help training, but they cannot create independent biological validation of unmeasured function. Abramson et al., 2024, Jumper et al., 2021, RCSB structure methods.

Biological virtualization means reproducing specified biological behaviour and causal dynamics in a digital model within independently tested conditions and error bounds. Invented biology presented as faithful reproduction is hallucination. A disclosed approximation can still be a useful scientific model. Missing validation does not prove a prediction false; one correct guess does not validate its mechanism or general reliability. The dice analogy concerns unknown reliability, not equal odds or an assertion that learned predictions are random.

7 Recording Access, Context and Stable Signals

The proposed measurement chain has to survive three questions. Access it: can we obtain the nervous-system observations relevant to a defined question, and what burden does access impose? Ground it: can we relate those observations to independently measured body behaviour on known clocks, in stated quantities and

with stated uncertainty? Keep it: does the recording and its mapping remain useful as tissue, movement and signal quality change? Keeping the recording usable is a different endpoint from a trainee retaining a skill.

Sources need organism, subject, region, task, modality, units, sampling and acquisition conditions. Cortex, spinal cord, peripheral nerve, muscle and movement are different observables. Electroencephalography (EEG) measures electrical activity using scalp electrodes; electrospinography (ESG) records signals over the spine; electroneurography (ENG) records nerve electrical activity; electromyography (EMG) records muscle electrical activity. EMG is not direct peripheral nerve recording. Separate studies of these signals do not become one synchronized voluntary cortex–spine–periphery chain by placing their charts together.

The resources associated with HHMI/Janelia, Allen, Harvard and other groups contribute particular observations to particular questions. Their value depends on that match. A broad catalogue cannot supply a paired signal that was never recorded; a model can estimate the missing value, but the estimate remains a prediction. An independent measurement can check it, but must remain a separately identified observation. When data are combined across organisms, source identities and physiological differences must remain traceable. Whole-body coverage is the objective to investigate, not a property conferred by the catalogue.

8 Physical Constraints

MyoSuite supplies muscle-based simulations with contact and task objects, making it possible to examine how a command becomes movement. The proposed chain keeps the neural predictor, goal, muscle controller, body mechanics and outcome score distinct. If the predictor selects the wrong key, a competent controller can execute that mistaken goal perfectly. Setting the joint directly to the desired pose would bypass the muscle-control question and must not be reported as a learned muscle action. These controlled computational studies still need their own tests of participant-specific biological accuracy. Caggiano et al., 2022.

NeuroTranslator names the implemented software joining particular recordings, models, replay and computer-body checks. NeuroTranslation Network names a proposed design for joining further components while keeping their origins and meanings clear. Neither name establishes full nervous-system coverage. Model size, learning from neural recordings and learning through simulated rewards require matched comparisons on the actual task and available hardware. A larger model or an impressive reward does not replace missing measurements.

Reaching, contact and release offer possible task relationships to compare across organisms. The mechanics still belong to each organism: a hand, paw and whole-body quadruped have different proportions, joints and control requirements. An averaged anatomy cannot establish a mapping that works for any of them. Physical scale, computational detail and statistical aggregation must also remain separate; a small organism can have a detailed model, while a pooled error can hide a failed recording. Biological generalization requires a test in the organism and conditions named by the claim.

9 Current Study: Neither Model Passed

The current hand study puts that demand for comparison into practice. Does retaining recent neural history improve the stated prediction without violating the other required conditions? Finishing a training run cannot answer it; the original conjunction of tests must decide.

The current development study comprises six sessions from one macaque: four training dates and two development-validation dates. Ninety-six recording contacts in the source's HandM1 motor-cortex region contribute threshold-crossing-rate (TCR) and spiking-band-power (SBP) measurements, counts of detected signal crossings and power in an archived signal band. They give 192 features, not 192 independently located electrodes. Its targets are two source-normalized finger-sensor groups: index and combined middle/ring/small fingers. They are not measured joint angles, forces or a full-body state.

Eighteen runs saved 720 epochs, passes through the training material. Both gated recurrent unit (GRU) candidates, sequence models with a mechanism for retaining earlier information, fail their original model gates. Both fail the required no-harm and positive neural-information conditions: the predeclared comparisons must show the required absence of harm and contribution of the neural information. All five separate history, order-use and auxiliary-benefit claims also FAIL. Reduced aggregate error relative to a selected ridge baseline, a simpler predictor trained with a penalty on large fitted coefficients, does not override a failed required combination of criteria. The negative report records these failures; it does not establish that either model is effective.

The failed comparison also leaves a reproducibility problem. Both candidates use Torch automatic differentiation to calculate derivatives and Adam to update the model from those derivatives and its stored state. Agreement between candidates using that common route cannot independently certify the route. Full independent NumPy calculation using a distinct numerical software library remains unverified, as do complete model/update state, random-number-generator (RNG) state, reload and training-trajectory evidence. The final-checkpoint/update gate therefore remains unresolved.

A continuation would need its own verified execution clearance, which is also currently unverified. The four reserved dates remain CLOSED. This article reports no new fitting, optimizer execution or reserved-test evaluation, and neither an accepted model nor full-training graphics-processor (GPU) promotion. The saved failures remain the result to explain. Programme computational evidence summary.

10 Earlier Studies

An earlier human finger-attempt classifier obtained 116 correct predictions among 150 later trials after calibration using the first 50 labelled examples of each recording session. In its separate keyboard replay, 41 of 60 source-labelled right-finger trials produced the intended contact, while all 54 supported commands actually issued were executed; 13 presses were unintended or mistimed. The controller's execution score

does not fix a prediction error. The model waited for a completed window; its earliest decision was one second after the cue plus computation. Inspected trials become development material if later tuning uses them.

A separate goal-driven simulated policy completed 32 of 32 scored phrase cases: eight reserved simulated orders crossed with two tempos and two poses, not 32 independent organisms. It uses a constrained generic four-key hand whose wrist, thumb and sideways finger-motion joints remain braced. It receives requested keys, phase, duration and smooth fingertip goals; displayed neural clips do not drive the simulated hand. The record preserves an interrupted original evaluator and a completed independent replay check; no test was reopened. Soft simulated key limits are not guaranteed physical travel bounds. A current saved development replay contains 336 frames at a 20 ms control interval, with 22 of 22 correct strikes, six of six chords, 22 of 22 timely releases, 34 ms worst timing error and no extra strikes. These are bounded simulation/readout results, not biological teaching, neural policy learning or unexpected real-world musical mastery.

The human recordings, macaque hand mapping and separate human EEG/ESG/ENG/EMG clips have distinct acquisition conditions and clocks. They do not demonstrate a voluntary central–spinal–peripheral pathway or drive a combined macaque/human arm simulation. Actual joint and force ground truth and physical unit calibration remain gaps. A task label, a normalized sensor value and a simulated joint angle must retain their different meanings. Programme computational evidence summary.

11 Failure Analysis

The failure narrows the next question: where did the extra error come from? In the reported index-finger comparison, the auxiliary decoder’s excess mean squared error, the average squared prediction difference, separates mainly into a mean offset and a smaller difference in residual variation. That decomposition locates a mismatch without identifying its cause. Trial composition, recording changes and state-dependent offsets could each contribute. Re-centering the validation predictions after inspecting that miss would use the answers to repair the prediction, so the adjusted result would no longer preserve the independent test.

Training-only context summaries show variation within trials, between trials and between dates; correlated overlapping frames are not independent observations. Named scalar calculations have bounded independent checks, including a preserved rejected skewness calculation, a measure of distribution asymmetry, and corrected arithmetic. Those checks do not demonstrate physiological state, neural drift or a useful new decoder. The proposed small context branch remains unexecuted. Any next fit requires independently checked training-only selection, normalization, controls, initialization, uncertainty and full update/state evidence, while preserving every original threshold. Current evidence and limits.

12 Viewer Capabilities and Missing Measurements

A graphical user interface (GUI) should let the reader tell a measurement from a prediction, a simulation from a missing value. Saved replay must preserve the source-time clock and complete-run metrics, even when the reader pauses at a persuasive frame. Cache-only playback displays an identified saved result without silently starting a new experiment; an uncached selection must show absence, and a failed source must clear stale geometry and results. Natural-language goals should become explicit task specifications. Receiving files in a chat does not itself establish inference or body control, so the interface must show what operation the system actually performs.

A saved replay and a live research view therefore need separate checks. The reader should be able to see which one is available, which source it uses and which measurements are missing.

12.1 What the online and offline versions require

The online candidate needs its whole directory. A disconnected reader instead needs the separate static export compiled from the original source; research views that use the application programming interface (API), the interface to the software service, are unavailable offline. Neither route has current whole-interface verification. Earlier bounded replay checks cannot establish that the present interface or the separate export works.

Reproducibility record. Moving opaque resources from the single HTML web-page source file into the package reduced that file from 113,298,857 to 34,295,597 bytes; its assets remain required for whole-directory online delivery. The separately compiled original-source offline export remains a different artifact. Browser verification was unavailable, and the candidate remains unverified and unpromoted. These are packaging measurements: they establish no interaction or loading latency, memory, crash-resistance or GPU benefit and no newer browser qualification. Computational evidence and limits.

13 Anatomical and Functional Validation

A viewer can expose an error only if the body it shows is itself accountable to references. Begin with proportions, muscles, bones and joints, then trace the neural routes and check contact and viewing geometry wherever those features are claimed. Each component and the assembled scene need lifelike three-dimensional review against a series of real-world references. Biological identification colours, useful close-ups and tilted orthogonal views can make those comparisons legible. Transparent layers, cross-sections and paired views can reveal relationships; illustrative signal paths must remain distinct from physical parts. This review tests the anatomy we show. Hidden physiological function needs its own evidence, however plausible the

image. A conceptual schematic remains an explanation, not a biological acceptance result.

A template surface or atlas provides a reference. Registration has to establish its correspondence to the recorded individual's anatomy before a location in the template can be attributed to that individual. The named source-binary reconstruction matches template geometry and atlas values, and its optional region boundary retains disclosed surface-mesh defects. Those bounded correspondences neither localize the recorded neurons nor establish their function. The retained framework illustration is historical and conceptual; its labels describe earlier computational work, not the current model-gate outcome.

The proposed left and right tripod views of the hands and arms need calibration, occlusion checks and independently verified timing. The motion-capture relay-composite would align or stitch those views. A separate perpendicular LiDAR view, using light to measure distance, may make particular surfaces easier to observe; it is not guaranteed to reveal every surface or solve neural registration. The acquisition method must earn its reconstruction claim against real measurements.

14 Proposed Skill Tests

Once the movement and its measurement can be trusted, the next task should expose something the present test cannot answer. Key presses separate selection, contact and timing errors. Threading adds flexible-object geometry, friction and corrective contact; puzzle manipulation adds object state and coordinated grasping. Archery or breakdancing needs forces, balance and wider body coverage absent from the current hand study. This gives the task ladder a reason to grow without pretending that a classifier or successful replay has already climbed it.

Reserve new task orders, object properties, starting positions and recordings before selecting the method. Then give simpler controllers and copied-example methods the same information and practice, and score the complete task: success, unintended actions, delay and failure. A selected clip cannot answer that comparison. Right-angle and non-right-angle mazes could test orientation and memory demands, but their difficulty needs measurement; 10× or 100× labels do not provide it. Computer training duration likewise cannot establish biological learning duration.

15 Development, Twins and Disease

The body also changes while a mapping is being learned. The native-development branch asks how recording access, maturation and changing function affect that mapping. Cook and colleagues provide species-specific skeletal staging in the fat-tailed dunnart. It does not establish corresponding timing in antechinus, a safe neonatal neural interface or an artificial rearing system. Armstrong and Ferguson studied age-related skin-wound scarring in *Monodelphis domestica*; skin healing does not certify scar-free neural access, spinal repair or stable electrodes. Cook et al., 2021, Armstrong and Ferguson, 1995.

Identical-twin, non-identical-twin and unrelated comparisons would ask which sources of variation affect a mapping. Similar genetics leaves differences in experience, physiology and recording location to be measured. Each comparison needs matched tasks, developmental stages and acquisition conditions before a channel or skill-transfer claim can be interpreted. The twins model must undergo component and assembled-scene reference review as well; kinship and a convincing linked image supply neither functional evidence nor visual acceptance.

Human-cell and graft branches ask defined questions about tissue contribution and host interaction. Revah and colleagues found maturation and circuit integration of transplanted human cortical organoids, laboratory-grown tissue representing aspects of the brain's outer layer, in rats. This does not make the whole host human, transplant expertise or demonstrate injury repair. The later image-panel correction is retained in the citation record. Native development, altered development, mixed-species tissue, grafts and disease conditions are separate proposed comparisons, each requiring its own evidence. Revah et al., 2022, author correction, 2026.

A disease experiment must identify the impaired function and the change produced by the interface. Compensation for interrupted motor output, improved task performance, retained learning and tissue repair require different endpoints. The cause of one condition may leave a route usable that another condition disrupts, so a benefit cannot be carried across them without its own test. These branches remain proposed; the present computational record demonstrates none of them.

16 Next Experiments

The failed comparison leaves us short of a qualified predictor, before we even reach the teaching question. The measured targets are still two normalized finger-sensor groups, without calibrated joints, forces or a complete body state, and neither recurrent candidate is admitted. A constrained controller could be studied by declaring its observations, permitted actions and feedback. If it receives explicit goals, its result must remain a goal-driven result; it cannot stand in for a neural prediction or evidence of biological learning. Only with those contributions separated can a trainer comparison ask whether useful movement assistance leaves lasting capability.

The next comparison can start with the grasp itself. The proposed two-person movement-only study asks whether pretraining on a qualified source person's movement improves a target person's movement forecast. Compare target-person pretraining with source-person pretraining under deliberately incorrect correspondence and under the proposed correspondence, matching architecture, initializations, update budgets, target-training batches and final-checkpoint selection. The proposed shared representation contains 13 movement features and 13 missing-feature masks: wrist-relative index, thumb and object sensor positions, digit aperture and causal wrist velocity. Proper wrist frames and a characteristic sensor-distance scale fitted only on each person's training observations separate coordinate orientation and size; a common 0.1 second reference interval also makes the velocity features dimensionless. The scale is not a measured bone length. Frames, units, clocks, gaps and sensor calibration still need qualification before this representation can be used.

Both homologous and mechanically analogous features remain eligible for the wider programme. This particular comparison tests a declared human sensor-landmark correspondence, not cross-species neural equivalence. Motion from another recording can supply a learned movement constraint without supplying a neural observation paired to the target person's action. Scans may constrain a specified anatomical gap, but an external scan or simulated movement cannot supply that missing measured neural pairing. Keep acquisition parents, overlapping segments and derivatives in one role before windowing or fitting statistics; development observations must not influence training scales or become independent final confirmation. This comparison remains unexecuted. Any later neural adaptation must use actual paired observations and retain its distinct input sets, causal timing, controls and protected partitions.

A trainer experiment needs supported interfaces and new evidence. Compare experienced, inexperienced and artificial trainers with matched task information, practice, assistance budgets and relevant burden. Include sham, delayed and unrelated-feedback controls. Improvement while guidance is present does not establish retained learning.

Withdraw trainer information while keeping any necessary movement bridge, practice, feedback and task access constant. Then measure delayed performance and unfamiliar variations against matched controls. Record the help that remains. If a gain fails to persist, that failure counts against learning. A retained gain still cannot establish tissue repair, restored pathways or regeneration: those claims require separate biological evidence.

17 Biological and Artificial Trainers

Brain2Brain names specified information links between nervous systems. Pais-Vieira and colleagues linked recorded activity and stimulation to share task-relevant information between trained rats. Prior training and dependence on stimulation limit what the result means: it is not instant acquisition of another animal's general skill repertoire. Pais-Vieira et al., 2013.

The proposed biological-trainer experiment compares experienced and inexperienced partners with matched practice. Genuine guidance, sham guidance without the intended information, delayed guidance and matched feedback unrelated to current need test different explanations. A simple artificial controller tests whether the neural trainer adds benefit beyond a less elaborate route. Random assignment, recording of actual help and independently scored outcomes are needed to distinguish expertise, information, timing and additional exposure.

An artificial trainer would supply guidance from a model trained on supported observations. Its comparison begins with what the model receives and controls. Desired goals, expert trajectories, privileged labels and sensory feedback give it different resources; supplying the correct action removes the prediction problem that a neural decoder is supposed to solve. Trainers therefore need matched task information, practice and relevant burden. A useful gain while guidance remains present can support an assistance claim even if no lasting learning follows.

18 Trainer-Withdrawal Tests

Test retention by withdrawing trainer commands while holding any necessary compensatory movement bridge constant. A bridge lets a trainee express action despite an interrupted pathway. Remove that bridge together with teaching and movement may become impossible regardless of learning: the test would confound body access with skill. Keep movement access, feedback and task conditions comparable between the baseline and later test.

Measure immediate performance, delayed retention and unfamiliar variations separately. Record how much help remained, whether task performance improved without trainer information, and whether the benefit exceeds matched controls. If the gain disappears when trainer commands stop, report assistance. If a gain persists, test its domain and uncertainty. Tissue repair still requires separate biological evidence; a retained task gain or successful compensatory interface cannot establish regeneration.

BrainNet is a network metaphor for specified links. It is not an operating market for biological mastery. Its value would have to appear in what a trainee can do through those links. Broad enablement and possible skill services motivate the programme, but they do not establish deployability, ownership of biological information or superiority over every other method. The proposal must earn support through its outcomes.

19 A grounded conditional learner with distinct output interfaces

The retention question now needs a mechanism: what information would guide the movement, and what independent observation could show that the guidance was useful? Follow one action through the proposed measurements and learner before adding a writer or attributing a gain to biological teaching.

Start with a movement we can measure well. For the proposed snap showcase, the arm begins extended with the palm down. The elbow and forearm swing and turn; the wrist and hand turn upward during the snap, then move down to rest. The model must account for that whole movement, including the preparation, contact and return. Finger pairing, organism-specific anatomy and timing still need a real-reference series; a human-looking animation cannot establish that a particular monkey can perform the same gesture.

The cup reach supplies a simpler precision comparison: relate recorded signals to measured movement before attempting the coordinated snap. Each task still needs its own anatomy, contact, timing and independent comparison.

That comparison connects three interacting groups: task-linked brain observations, named-muscle observations and a musculoskeletal simulation. A learner would relate the observations to the mechanics; independent movement and contact measurements would test its predictions. EEG records scalp electrical activity and surface EMG records muscle electrical activity. A separately recorded nerve or spinal

signal needs its own declared channel. The proposed apparatus uses two arm sleeves, scalp EEG and synchronized, calibrated left and right tripod views. Vendor, channel, bandwidth and clock qualification remain open.

Intent to action is the target. By understanding and measuring the intermediary, we aim to make that mapping as close to one-to-one as possible within a declared task. Specify what counts as the intended action, the participant and session, state and context, permitted history, tolerated error and independent reference. The test then asks whether observing the middle reduces ambiguity and improves prediction or qualified control. It does not assume a unique hidden intention or claim that the best possible mapping has already been achieved.

The learner's first test would compare its predictions with recorded brain and muscle activity and with measurements of where the hand moved and when. Sensory feedback and physical action would add a further question: what did each separate output interface actually deliver? A successful prediction cannot answer that question, and EEG and EMG recorders alone cannot write sensation. For this proposal, no paired trial, trained model, delivered sensory output or physical actuation is established. Sources and permissions, physical and computational behaviour, clinical interfaces, reference comparisons and held-out tests still require independent assessment.

The retained LINK comparison belongs to a different experiment. Its 18 runs and 720 saved epochs are not paired EEG/EMG/movement trials; both model-admission gates and all five benefit claims failed. The prospective methods below would have to supply the missing paired evidence and separately test recording, prediction, simulated movement, sensory delivery and physical action. A result for one of those jobs would not establish the others.

19.1 Form, function and action need a measured middle

A structural map constrains what a system could do. A NeuroConnectome maps specified neural connections; its organism, coverage, resolution and source would have to be declared. The signals passing through those connections still have to be observed or inferred, with the difference made explicit. To predict the next action, we need the processes that turn structural constraints into an outcome. The proposed programme follows form to function to action, while intent to action names the task-level translation we want to improve.

DNA to RNA and RNA to protein put a process between a starting constraint and its downstream product. Regulatory context, RNA processing and cell state change what that process produces; protein structure alone does not determine the behaviour of a whole cell. AlphaGenome, RNA or cell-state models and AlphaFold illustrate these different modelling jobs. AlphaTranscriptome names a desired predictive capability here. Connecting the predictors would still leave system dynamics and the environment to account for, with experimental outcomes needed to test the combined prediction.

AlphaGenome regulatory prediction [1]; scGPT cell-state modelling [2]; AlphaFold 3 structure prediction [3]. These provide modelling examples, not evidence of an integrated cell programmer. Our account of AlphaGenome and scGPT is limited to their

reported modelling roles; their complete methods have not been independently assessed here.

In the biological pathway, Brain2Spine and Spine2Body name proposed links to test; their composition is the Brain2Body aim. The spinal cord integrates descending and sensory signals and contributes local coordination, reflex regulation and adaptation. This is a different mechanism from RNA processing, but it serves the same explanatory purpose: account for the context-dependent process between an initial constraint and its outcome. Supported spinal observations would test that account. Where they are absent, the internal states remain inferred or simulated. Physical spinal work remains read-only.

Spinal sensorimotor adaptation [4] is a constrained example, not evidence of unrestricted skill transfer or validation of our proposed pathway. Our comparison draws on the abstract and relevant author-manuscript passages; equivalence to the final published edition and the complete methods and statistics remain unverified here.

Thought2Action relates a declared intention to an independently measured action. Action2World follows the action to its consequence in the task environment; Thought2World names the proposed composition. Body2World drew attention to the consequence outside the body; Body2Action focuses on body mechanics and a specified task outcome, while Brain2Motion focuses on the movement pathway. These working labels identify related scopes for a test. They do not establish interchangeable or working controllers.

The engineering route connects neural activity to musculoskeletal simulation and then to a programmable model, toward NeuroTranslation. The proposed movement-focused learner is a large movement model. System 3 thinking names the proposed human–AI supervisory partnership around it. The supervisory model would compare task strategies and diagnose where neural or mechanical predictions miss independent measurements, then propose training or calibration comparisons for the signal mapping and virtual body. The task learner would perform the declared prediction or control; the supervisor would help choose what to test. Neither role replaces the biological spinal intermediary or a physical interface. Final held-out outcomes must remain outside rewards and model selection.

Shaw and Nave's SSRN preprint, *Thinking—Fast, Slow, and Artificial*, and its institutional discussion provide conceptual context for this proposed human–AI role. [5] The comparison is limited to that framing; the complete preprint and institutional article have not been reviewed here. Neither establishes a third anatomical brain system or validates this programme's proposed learner or supervisor. Expert footage may supply movement and tactical examples, with rights and camera uncertainty accounted for; it supplies no measured expert EEG, EMG, muscle force or stimulation plan.

Action changes the outcome. Its sensory consequences inform perception and the next intention. Independently measured movement would anchor the action part of that loop; it would not by itself decode a unique sensation or perception. Known stimuli, measured feedback and appropriate recordings or reports would have to test those further mappings. The return path must therefore be measured rather than drawn into existence. The full Interoceptomics aim includes internal-body sensing and connectivity; external touch and proprioceptive tasks contribute evidence but do not

by themselves solve that wider problem. A prediction, assisted execution and retained skill without assistance answer different questions. Each wider claim must earn its own independent comparison.

19.2 Context connects. Grounding checks

Within a reach, context has a concrete job: relate brain activity to muscle activity in this task and at this time. Attention is one proposed way to represent those relationships. The model compares a query with keys and uses the resulting weights to combine their values. Masks keep unavailable observations and future samples out. The calculation weights information; it does not establish that attention understands all context.

Vaswani et al. (2017) [6]; Tsai et al. (2019) [7]. These methods motivate the computation; validation of the proposed paired system still requires the independent observations below.

A pattern can fit inside the model and still describe the wrong movement. Grounding therefore means testing the declared prediction against qualified observations under the same conditions. Independently measured movement would check the outcome, while muscle activity would add an observation that constrains the brain model. Even a matching reach cannot identify a unique private experience. No acquired paired data or trained model is established for this proposed loop; those comparisons still have to be made.

19.3 Muscle activity tests the brain model

Record brain and named-muscle activity during the same reach. In the proposed paired trial, EEG would sample scalp electrical activity and surface EMG would sample activity associated with the named muscles. Paired training could use the muscle observation to constrain their relationship. Then withhold the muscle target and movement reference from the EEG-only predictor: can the permitted brain observations predict a specified muscle-activity observable or movement?

Pairing and calibration remain proposed. Shared cues and movement artefacts could give that predictor a shortcut, so the comparison needs controls for both. The muscle observations would cover only the named muscles, and EMG supplies neither afferent nerve activity nor direct force. Sensory claims still need known stimuli and measured feedback. Several neural states could remain compatible with even a well-matched pair; the pair could narrow a task-specific explanation without revealing the complete hidden brain. Checking the movement itself requires the next independent observation.

The candidate reference at https://www.nature.com/articles/sdata2018133 still needs a check of its exact support and reuse rights.

19.4 Measured motion tests the virtual body

Measure the same hand and arm from the aligned left and right tripod views. The proposed motion-capture relay-composite would align the two point maps in a common clock and coordinate frame, then join them into one composite while retaining each point's origin. Before that joined map can check a prediction, its calibration, alignment and capture precision would have to be independently verified.

The movement map would constrain the external movement model: surface motion and pose in the task's physical frame. To explain that motion mechanically, the internal musculoskeletal model would describe specified bones, joints and soft tissues. Organism, developmental stage and task would determine the appropriate anatomical sources. X-ray projections could inform bone and joint relationships; appropriate MRI references could inform specified soft tissues. Registration, rights and source-specific limits would need independent checks. No scan set is selected for this proposed comparison. Anatomy still leaves activation, forces and neural function to test.

With a qualified composite, the musculoskeletal-simulation physics-engine's joints, muscles, contacts and physical limits would face a measured comparison. Fit the model on training data, then compare its predicted reach with held-out measured movement. A force claim would require an additional force measurement. The body's form limits possible motion; measured function would test the mechanics and show where they need refinement.

Seth et al. (2018) [8]; MyoSuite documentation [9]. The simulation tools supply methods for that comparison; the proposed capture and body models still need its independent validation.

A lifelike hand makes that comparison visible. It cannot establish biological fidelity. Capture alignment, precision, mechanics and biological fidelity have not been demonstrated by this proposal, and captured geometry cannot establish hidden neural state, internal forces or unique subjective experience. The measured reach would check a movement prediction within those limits.

19.5 Two pairs. One conditional learner

The two comparisons would meet in the proposed large movement model: EEG observations paired with muscle activity, and measured movement compared with predicted mechanics. Large movement model names the proposed family; deep neural network describes a possible implementation. Large refers to the intended scope, rather than a demonstrated parameter count. Encoders would receive only the observations permitted for the named test. Two branches would summarize the relationships and meet in a shared task representation; separate output parts would predict each direction's targets.

Vaswani et al. (2017) [6]; Tsai et al. (2019) [7]. These computational methods offer precedents for a design, leaving the proposed paired learner and its performance to be tested.

Signal features belong to branches, and branches belong to the learner. That hierarchy describes the computation, not two anatomical thinking systems. A common clock,

task and physical frame would connect the branches; independent measurements would check their errors. Form constrains the prediction, and checked function would guide refinement. Several hidden states could still fit the same reach, so neither a shared representation nor conditional predictions would establish an exact inverse nervous system. The architecture has not been selected, implemented, trained or independently accepted. To choose it, we would have to compare it with simpler baselines on appropriately reserved observations.

19.6 Learn the movement, then test what transfers

A reach can provide a training target even when no EEG accompanies it. Earlier movement supplies the input; a later observation supplies the target. A separate proposed comparison would ask whether learning that relationship in one person helps forecast another person's grasp-and-lift movement. The WAY-EEG-GAL descriptor provides a relevant continuous-recording candidate, alongside its distinct windowed trials and derived lift summaries.[10] This movement-only question needs no AllLifts Run/trial join or invented cross-person neural pairing. The original condition-specific five-weight study remains a separate design.

The mapping has to explain what two bodies share. Homologous structures share evolutionary anatomical origin; mechanically analogous structures can serve comparable roles without that origin. A human digit aperture and a robot gripper opening could support a task-space comparison after separate mechanical qualification; this human-only proposal would not test that robotic extension. Joint degrees of freedom, axes and signs, proportions, muscle–tendon roles, loads, inertia, contact and time scales remain body-specific. Similar geometry cannot establish equivalent dynamics or neural coding. Retargeted motion remains derived; tracking, estimation, prediction and simulation keep their different evidential roles.

For the proposed two-person comparison, the representation contains index-, thumb- and object-sensor positions relative to the wrist, digit aperture and causal wrist velocity: thirteen features, with thirteen availability masks. Each participant needs qualified position units, a proper wrist-frame rotation, orientation convention and sensor-attachment calibration. A positive median wrist-to-index-sensor distance, expressed in qualified metres, supplies the characteristic length from assigned training support only; it is not a bone measurement. Multiplying normalized velocity by the same fixed 0.1-second reference interval makes all thirteen features dimensionless. Physical velocity errors invert that scaling with length divided by 0.1 seconds. Missing features receive no target loss; an unavailable observation is not a measured zero. Full finger joints, individual muscle forces and subject-specific inertia are not supplied by these sensor features.

The continuous target would be the same observed feature set 100 milliseconds beyond the latest actually available filtered input time. The proposal retains fifty frames at 100 Hz and a causal 51-tap filter before decimating qualified 500-Hz kinematics by five. Its 50-millisecond linear-phase delay must remain in the physical timing account; shifting information backward would change the prediction claim. Coefficients, passband, timestamp convention, gaps and boundary handling still

require qualification. Only named kinematic columns may enter: the documented kinematics matrix also contains force-plate signals.

Identity and independence answer different questions. Trustworthy versioned participant-release attribution, an immutable selected archive entry and a qualified capture can identify the participant record and input object without requiring one particular internal ID-field spelling. They cannot establish biological identity or independent original acquisitions. Shared samples, trims, derivatives and overlapping recordings must form indivisible acquisition groups before filtering, windows, normalization or role assignment. Different names, checksums or series numbers do not prove separation. The complete public non-ST candidate roster and these relations still need establishing; all 294 rows of the prior AllLifts table remain excluded from training, evaluation and assigning trials or acquisition groups, and ST, competition storage and the reserved LINK recordings remain unavailable.

Qualified P1 groups would supply source pretraining; P2 would supply target training and development. With $G \geq 2$ independently established P2 acquisition groups, the fixed identifier-hash ranking assigns the first $\lfloor 3G/5 \rfloor$ to training and the remainder to development. Canonical original-parent identifiers, their serialization and the roster must be fixed before ranking or examining outcomes. This is a nominal 60/40 split, not chronological ordering or a promise of condition balance. If unresolved ancestry merges P2 into one group, the comparison stops; splitting its windows cannot manufacture an independent development group. No group or segment has been assigned by the proposed identity method.

Three matched arms would test the transfer. B0 pretrains on correctly mapped P2 training movement. B1 and B2 use identical P1 windows and order: B1 sets the index-role input to the thumb sensor, the thumb-role input to the object sensor and the object-role input to the index sensor, while B2 retains the correct correspondence. Both share the correct source calibration and scale before that corruption, keep the wrist anchor, and use the same correct target adapter and ordered target batches. Parent-balanced sampling must record unique windows and repetitions. The false mapping can be relearned or can alter feature availability; matched update counts do not match source diversity. These are limits on attributing any difference to anatomical correspondence, not reasons to replace the control after seeing its result.

The proposed learner is one GRU layer of hidden width 32, with 26 inputs and thirteen outputs. Each arm and fixed seed—17, 31 and 73—receives 200 pretraining and 100 target updates, batch size 16, identical per-seed initialization and the fixed last checkpoint. Adam uses learning rate 0.001, betas 0.9 and 0.999, epsilon 10^{-8} and zero weight decay, without clipping, scheduling or early stopping. Training minimizes mean squared residuals over observed target features; a batch with no observed targets is rejected. The resulting 2,700 updates are planned, not performed. No external checkpoint is selected; numerical updates, filtering, precision and actual workload limits remain to be checked independently.

For each development parent and seed, the proposed score takes the square root of summed squared errors divided by the count of observed target features, then averages parent scores without weighting by parent duration. A zero denominator rejects the comparison. Report feature counts, missingness, physical-unit errors, uncertainty and persistence at the same information endpoint, including every worse

parent or feature. B2 would have to lower normalized error against both B0 and B1 in every fixed seed, with no applicable original criterion failing, to justify another investigation. This remains a two-person development comparison, not independent final confirmation, population transfer or biological teaching.

Broader motion collections could supply additional movement examples only after their exact assets, rights, task coverage and original-recording overlap are qualified. In AddBiomechanics, measured markers and ground reactions must remain distinct from fitted kinematics and estimated inverse-dynamics torques. AMASS requires checking the selected asset's research-use, non-commercial and redistribution restrictions. Copies across collections stay with their original acquisition. LaBraM's WAY source-family exposure is flagged, while exact checkpoint/participant/trial exposure remains unknown; reserving a recording later cannot erase earlier pretraining exposure. No such checkpoint is selected for this comparison.

Before adding X-ray/CT or MRI, inspect the anatomical sources already used by the selected musculoskeletal model. A scan is useful when it constrains a stated gap or supplies an independent check beyond that model. Its person or species, scale, registration, rights and uncertainty still matter. Another person's scan remains an external anatomical reference; a generic model does not implicitly measure this participant's geometry, activation or forces. No scan set is selected here.

Actual source eligibility, acquisition groups, units, frames, clock and calibration remain unresolved for this continuous comparison, so no new fit or successful transfer is reported. A useful movement prior could later be tested with actual paired neural/body observations and the permitted input masks. Its transferred targets would still not become simultaneous measurements. The next result must identify where the mechanical forecast improves, where it worsens and which measurement or assumption accounts for the difference; neural equivalence, causal physiology, sensory delivery and retained skill would require further experiments.

19.7 Pretrain. Pair. Verify. Refine

Those comparisons start with the observations we may use. Data rights, calibration and held-out subjects, sessions and tasks determine which can enter the study; resolve them before processing data or choosing a model. Open-source code does not grant blanket permission to use datasets, checkpoints or clinical interfaces. Qualified data could first train representations for each recording type. LaBraM is one EEG encoder candidate, not a complete brain-body writer. Synchronized paired observations could then train the declared mapping from permitted inputs.

Paired prediction would give later refinement a baseline to beat. Before optional reinforcement learning, post-training or task-specific fine-tuning, the physical predictions and reward need checking against independent outcomes. Offline reinforcement learning needs qualified coverage of actions, transitions and rewards: labels alone cannot tell us what an untried action would do. Backpropagation can update differentiable neural components, but it cannot automatically pass through every physics engine; universal differentiability is not established. Simpler baselines and frozen final tests remain through every phase.

Jiang, Zhao and Lu (2024) [11]; Kumar et al. (2020) [12]; Seth et al. (2018) [8]; MyoSuite documentation [9]. These sources offer candidate methods for the sequence; validation of the proposed paired system still requires the experiment below.

No data acquisition, training, fine-tuning, RL rollout or accepted checkpoint is established for this proposed loop. The next experiment must test what the prediction and reward represent against independent outcomes. A reward that the computer can calculate supplies no biological proof on its own.

19.8 Respond quickly. Improve carefully

The proposed response path would use a frozen accepted version to estimate the current state, predict declared targets and prepare encoded outputs for a separately qualified interface. Permitted causal time windows would limit it to the allowed past and present observations. Estimating the state of this moment would not change learned weights. A separate adaptation path would inspect errors, calibrate or recalibrate, and propose a candidate update; independent evaluation, a restorable checkpoint, restoration proof and applicable authority would precede promotion. Examining errors would not silently change the running model, and no unchecked live weight promotion would occur.

The fast-and-slow comparison is a Kahneman-like analogy for this proposed division of work. It identifies neither two biological or anatomical systems nor inherent thinking speeds for EEG, spatial data or other sensor modalities. Real-time performance would have to be measured across acquisition, filtering, alignment, inference, output preparation, transduction or actuation, feedback and rendering. Report delay tails, jitter, missed deadlines, uncertainty and coexistence with healthy workloads; component inference time alone cannot establish end-to-end performance. No accepted model, running system that adapts under these rules or real-time demonstration is established here. The loop has no measured end-to-end latency or accuracy guarantee.

19.9 Write sensation. Enact action

The prediction ends at the computer. A separate interface would have to turn it into sensation or action. Input and output describe direction relative to that computer; afferent and efferent describe physiological direction relative to the central nervous system. A computer output could drive a sensory interface supplying input toward the nervous system, while a computer input could record activity related to the body's motor output. Both boundaries must remain explicit.

The sensory route would use encoders and a separately qualified visual, haptic or other sensory transducer to deliver specified feedback. Direct neural writing remains a separate later research interface, with its own device, measurement, safety and clinical evidence. Direct cortical writing could have a sensory role without being a peripheral afferent signal. Predicting an EEG-like pattern would not deliver sensation: the sensory content, interface, authority and actual measured response would need their own tests, including conditions, error and delay.

The action route would send separately qualified encoded controls through a controller to a virtual body, a robotic actuator or a later qualified biological interface. Action encoders, controllers and actuators have distinct jobs in that route. A robot command is an action-role output, not an anatomical efferent nerve. Simulated action, physical actuation, retained learning and biological causal claims each require their corresponding evidence. EEG and EMG recorders neither supply sensation nor actuate a body. These sensory and action subinterfaces belong within the unchanged five programme pillars.

Failed access to a clinical source supplies no methods evidence. Implanted precedents cannot establish that EEG or EMG recorders are equivalent to devices that deliver sensation or produce an action. The acquisition and virtual-modelling plan excludes physical sensory writers, transducers and body actuators; these additions remain unqualified and unpriced. Acquisition and virtual-modelling costs do not establish what delivering sensation or action would cost.

This recording-and-virtual-modelling proposal cannot test physical sensory delivery or biological actuation. No such physical interface, stimulation, acquisition or actuation is established here, and no loop, training or physical-writing result is claimed.

A physical study would need its own qualified interface and procedure. This proposal supplies no stimulation dose, implantation, electrode-placement, acquisition or actuation procedure. It authorizes no stimulation, participant acquisition, biological actuation, new equipment purchase or training, and grants no participant, clinical, device, purchase or deployment permission.

19.10 Close the loop with evidence

Return to the cup and test each prediction against the observation it was asked to explain. The brain-to-body prediction would face withheld observations; the body-to-recorded-neural estimate would face its own recorded target. A sensory or action interface would need a further test of what it delivered, under which conditions, with what error and delay. Two models can agree and still be wrong; a high reward can have the same problem. Neither agreement nor a successful prediction would, by itself, identify a unique inverse brain or establish someone’s private experience.

The programme’s distinction remains that approximation gets us towards plausibility and virtualization gets us to precision. Independent measurements, repeatability, causal checks and stated error bounds must earn that precision. The original held-out and real-reference controls would test the prediction; retained-learning controls would test whether a gain remains after useful guidance ends. Precision is not an automatic result, and the proposal establishes no perfect biology.

The next experiment must report where each declared prediction works and where it fails against its independent measured target. Its answer would determine which part of the proposed loop needs refinement.

19.11 Technical representation and mask

The information boundary must survive the calculation. A model given tomorrow's sample or today's withheld target could make a persuasive prediction of an answer it was never meant to receive. The representation and mask specify which comparison the proposed attention computation would actually perform.

Learned query, key and value representations give the proposed attention calculation a precise form. An admitted mask would restrict which observations may contribute; biological grounding still requires the independent comparison.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(QK^{\top}/\sqrt{d_k} + M\right)V.$$

Here Q, K and V are queries, keys and values; M expresses the admitted mask. The dimension d_k is the common feature width of each query and key vector. The equation is methodological context from the supplied technical proposal. Modality, channel, task and time are part of the observation contract. No weights, EEG encoder or gradient route are selected or executed.

19.12 Permitted observations and causal modality routes

Declare B=EEG, E=EMG, Y=independently measured movement, C=task/stimulus/context, and P=the source-bound physics model. Training may relate qualified paired observations. Inference uses only its explicitly admitted subset:

- EEG-only: permitted past/present B and declared context predict a specified EMG/movement target. Withheld E, Y and future samples cannot enter the predictor.
- Body-only: permitted E and/or Y predict specified recorded neural features as a conditional distribution with uncertainty. This is not a unique inverse brain.
- Hybrid: additional modalities require a separately named route and evaluation; a hybrid result cannot be reported as EEG-only.
- Simulation-derived inputs stay simulated. They do not replace independent observations or inherit the biological validity of a rendered body.

Separate training signals from evaluation references. Cycle consistency can make two models agree while both are wrong. Shared cues, movement artefacts, subject identity, session overlap and preprocessing leakage need the original controls. Do not expose final held-out targets to model, physics or reward selection.

19.13 Representation, supervised learning and policy refinement

0. Establish source rights, subjects/tasks, measurement units, calibration, uncertainty, clock alignment and frozen subject/session/task splits. Audit pretraining overlap before preprocessing and model selection.

1. Pretrain modality representations on the qualified inventory. LaBraM is one candidate EEG encoder: neural-tokenizer training, masked neural-code prediction and downstream fine-tuning. It is not a complete brain-body writer. Code, checkpoint, dataset and individual-use rights remain separate qualifications.
2. Learn paired supervised mappings with named targets and explicit modality masks. Compare appropriate simpler, cue-only and artefact baselines. Keep raw, processed and model-derived values distinct.
3. Qualify physical predictions and the reward verifier against independent measurements. A successful task, smaller calibrated movement error and valid physical states are prospective reward ingredients, not completed results. Declare loss/reward weights on training/development evidence. A computable objective or high reward does not confer biological ground truth.
4. Where admitted action/transition/reward coverage supports it, evaluate optional RL and post-training. Offline trajectory labels alone do not establish the consequences of actions never supported by the data. Conservative Q-Learning is a methodological candidate, not a safety or performance guarantee here.
5. Fine-tune for the declared task and qualify both directions on frozen final tests, uncertainty, baselines and every original threshold. Keep failures.
6. Serve a frozen accepted version. Propose later updates separately and promote only after independent acceptance, restoration proof and applicable authority.

An illustrative supervised objective would combine declared errors for specified EMG, movement and recorded-neural targets with applicable physical constraints and optional consistency terms. Each term needs its units, measurement uncertainty and target definition before a weight is chosen. Consistency can check whether the outputs fit together; an independent observation checks whether they fit the measured outcome.

Backpropagation changes differentiable components of the computer model; biological feedback is a different process. A physics engine in the path does not automatically give the learner a gradient. Qualify the actual OpenSim/MyoSuite rollout and optimization route, or a separately validated surrogate, rather than assuming gradients through every route. No training, RL rollout or task checkpoint is created or certified here.

20 A larger model still owes us a learning experiment

A model could predict a hand’s path more accurately while teaching its owner nothing. That distinction matters as the programme expands from recording-specific mappings toward a proposed Large Movement Model. The new hypothesis is that greater amounts and greater diversity of qualified neural, muscle and movement data could improve generalization. Brain2Body asks what any resulting prediction would let a trainee do, what information or assistance supplied the gain, and what remains when teaching ends. A scaling result would answer the first modelling question only within its tested domain.

20.1 Four training sources, five component roles

The expanded training proposal has at least four source groups: neural observations; muscle and other myographic observations; measured motion and displacement; and data generated by physics-based motion or musculoskeletal simulation. Suitable neural sources may span brain, spinal, peripheral and relevant visceral recordings. Their actual modalities, sites and task roles remain distinct. The present emphasis is motor–sensory. Myoception is the programme's term for the sensory side of musculoskeletal contraction and movement, including proprioception, tactile feedback and other task-relevant sensations linked to action. It marks proposed scope, without establishing a new external scientific standard. That scope supplies neither acquired coverage nor body-wide simultaneous observations, and the four groups are not an exhaustive taxonomy.

Known tactile stimuli, proprioceptive feedback, contact, force and task context contribute different evidence. Stimulus delivery, reported sensation, recorded sensory activity, calibrated force and tracked configuration cannot be substituted without a tested observation relationship. EMG records muscle electrical activity, not direct peripheral nerve or sensory activity; kinematics alone does not measure proprioceptive experience. Each measured, reported, estimated or simulated quantity keeps its provenance.

The simulation group supplies potential training examples as well as constraints: trajectories, state transitions, muscle–tendon or contact-force outputs and declared movement priors. Record the simulator, body-model version, parameters, solver, control representation and underlying sources. Joint-target control must not be called muscle-driven control. Retargeted recordings retain original lineage and transformations; generated states or forces remain simulated. The generator requires calibration against appropriate measured support with uncertainty and training-only role boundaries.

Test its contribution by comparing a fixed-real-data reference with a simulation-augmented arm under matched permitted information, compute and original criteria. Record real-example exposure, simulated exposure and updates in each arm; a fixed real corpus alone does not match training exposure. Count simulated amount and diversity separately from real acquisitions. Held-out real observations must check the result without entering generator calibration, tuning, reward choice or model selection. Wrong dynamics, contacts or source correspondence can make the synthetic contribution fail. Generated data is never independent biological ground truth for the mechanism that produced it. Generator methods, implementation and use conditions remain to be checked; no frontier implementation or dataset is selected here.

Five component roles describe another axis: brain signals, muscle signals, measured movement, a virtual body and a learning model. A virtual body can supply training data and physical constraints, while a learning model can draw from several source groups. Neither count excludes another useful, properly characterized observation. These resources give the later teaching study more hypotheses to test, rather than supplying the missing retained-learning result.

20.2 Breadth needs a controlled comparison

Amount and diversity must be varied separately. Repeated windows from one acquisition increase presentations without adding independent subjects, days or tasks. A proposed amount comparison would increase support from a stated source distribution. A diversity comparison would retain a declared amount while changing named aspects of that distribution, with a crossed comparison testing their interaction. Organism, subject, session, movement, body region, modality, apparatus, environment and measurement quality require separate descriptions. Dataset count alone cannot measure them.

The architecture, permitted input information, targets, preprocessing and selection rules would remain fixed. Source rosters, repetitions, updates and measured computing work must accompany the results; equal update counts alone do not establish matched compute. If diversity also changes recording quality or task difficulty, the comparison cannot attribute its outcome to diversity alone. All exact budgets, controls, seeds, uncertainty methods and decisions remain to be registered before a new experiment. This proposal changes neither the adopted continuous WAY-EEG-GAL development design nor any original protected partition.

A candidate amount–error curve, for a fixed diversity condition and evaluation domain, is

$$E(N) = E_{\infty} + A(N/N_0)^{-\alpha}.$$

$N > 0$ and its fixed reference $N_0 > 0$ use the same declared training-amount units. E is a chosen nonnegative error; E_{∞} and A use its units, while α is dimensionless. This decreasing-error candidate requires $E_{\infty} \geq 0$, $A \geq 0$ and $\alpha > 0$: $A > 0$ gives strict decrease, while $A = 0$ gives constant error. Positive α alone is insufficient. Predictions must also respect any upper bound of the metric throughout the declared fitting and prediction range. The floor is a prospective curve parameter, not an established irreducible biological limit. No exponent is estimated here and no universal law follows. Constant, saturating, increasing, non-monotonic and domain-specific alternatives need prospective comparison and predictive checks at withheld scaling levels. Observed negative transfer remains a result even when this constrained candidate cannot fit it. Related windows cannot supply independent uncertainty replicates. Neural-target, EMG-target, pixel and physical-motion errors retain their own units and denominators.

Generalization would be evaluated separately across excluded subjects, sessions, tasks, datasets and, where mechanically qualified, species. The target-only/no-transfer reference, applicable incorrect correspondence, simpler predictors and original null, shuffle, cue/confound and causal-time controls remain. Negative transfer, worse domains and saturation stay in the record. A better pooled score cannot discharge an original per-output no-harm criterion or erase known pretraining exposure.

20.3 Build a composite without inventing a shared observation

FALCON M1-A and AJILE12 illustrate complementary source candidates. The fixed FALCON M1-A release describes simultaneous M1 activity and processed intramuscular EMG from one rhesus macaque; released synchronized motion capture is not established. Its acausally processed EMG targets limit causal real-time interpretation.

The small minival subset is drawn from calibration data for submission-format checks and is not independent validation. AJILE12 describes intracranial recordings and two-dimensional, video-estimated upper-body pose from twelve clinical participants. Paired EMG and instrumented three-dimensional mocap are not established there. Its pose-label construction is participant-adapted; particular held-out-day estimator exposure and numerical alignment uncertainty remain unqualified. FALCON M1-A release; FALCON methods; Peterson et al., 2022; AJILE12 release.

Qualified source-specific pretraining could exploit each relationship. A transfer model would then have to demonstrate a correspondence on actual target measurements. The separate recordings never become one simultaneous neural–muscle–movement event. The release descriptions do not establish downloaded recordings or a trained model. Their exact use conditions, versioned files, integrity, original acquisition relationships, timing, units, earlier exposure and independent partitions still need checking before training. Retargeting between homologous or mechanically analogous structures also requires declared geometry, frames, scale, load, contact, timing and uncertainty. Similar movement cannot establish identical neural coding.

20.4 Model the inside and the outside of one action

When a hand closes on an object, external displacement and contact put demands on internal mechanics. The proposed world model would represent specified musculoskeletal states alongside the supported body, objects and environment. NeuroTranslate would connect named observations and targets; MoCap ReComp would provide its separately qualified source-preserving movement composite; MyoSim Replay would simulate declared mechanics. Their proposed combination is the Large Movement Model. Neither that name nor a neural composite implies a measured whole nervous system, equivalent cells across organisms, complete internal organs or unique access to private experience.

The model must distinguish measured neural activity, muscle electrical activity and tracked motion from estimated activation, internal forces, latent neural state and simulated motion. EMG is not direct peripheral nerve recording or automatically calibrated force. A proposed transition model could forecast a distribution over joint configuration, generalized velocity, muscle-model activation and environmental state from permitted past observations, context and declared controls. Separate observation models would connect each sensor to the state it can constrain. An unavailable observation stays unavailable; a conditional reverse prediction cannot identify a unique hidden brain.

Physical tests would compare aligned predicted movement with independent measured motion, and any internal-force claim with appropriate independent force evidence. Joint limits, contacts, loads, energy accounting, friction, gravity and numerical solver behaviour need their own checks. OpenSim supplies established methods for declared musculoskeletal models; it cannot confer biological validity on this proposed assembly. Seth et al., 2018. A rendering can show where a model says a muscle pulls. Only the applicable observations can test whether that pull explains the claimed result.

20.5 One muscle pulse, two simulated trajectories

A moving virtual hand has answered a mechanical question before it has answered a teaching question. In one numerical diagnostic, we compared two trajectories of the installed `myohand_r` human right-hand mechanism on a fixed passive torso. A muscle excitation was added to one branch. The other supplied its matched control. This asks whether the specified software input changes the computed state. It provides no neural decoding, measured biological response or retained-learning outcome. The mechanism used `myo_sim` 0.2.3, commit `93b0ca8f4ec90c9899ee7f05fee561e9911da91b`, MuJoCo 3.14.0 and NumPy 2.5.3.[13–16]

20.5.1 Mechanism and matched comparison

The compiled model has 23 scalar hinge degrees of freedom, 39 muscle–tendon actuators, 39 activation states and zero sensors. Both branches began with matching full recorded initial states, including positions, velocities and activation states. The control branch received no additional excitation pulse; the pulse branch received dimensionless excitation 0.2 at `FDS2_r`, with all other controls zero. Excitation is a muscle-model input, distinct from a joint-position request, measured EMG or neural activity. Gravity and passive dynamics remain present, so the unpulsed trajectory need not remain stationary.

The native Euler integrator advanced each branch through 100 explicit steps of 0.002 s, giving nominal $T = 0.200$ s. Pulse steps 10–59 inclusive correspond to the pre-step interval $0.020 \le t < 0.120$ s. Each branch received explicit initial and final forward evaluations to synchronize state-dependent quantities. Thus the record contains 200 explicit integration steps and four explicit forward evaluations. The executed source path also implies two compilation calls and one construction-time internal forward evaluation. Those counts do not enumerate every internal native solver operation.

The construction path changed physical parameters even with optional `inertia_floor=None`: each diagonal inertia component of four fixed scaffold bodies changed from 1 to 10^{-4} kg m^2. This native transformation is part of the tested mechanism, rather than evidence that all parameters remained untouched. No additional physical retuning or `implicitfast` fallback was used. Position, orientation, scale, contact behaviour and biological fidelity have not been independently calibrated by this diagnostic.

20.5.2 Recorded response

For pulse branch P, control C and the synchronized final state, the reported quantities are

$$\Delta q_\infty = \max_{1 \le j \le 23} |q_{P,j}(T) - q_{C,j}(T)|, \qquad \Delta a = a_{P,\mathrm{FDS2}}(T) - a_{C,\mathrm{FDS2}}(T),$$

$$\Delta F = F_{P,\mathrm{FDS2}}(T) - F_{C,\mathrm{FDS2}}(T).$$

The first quantity is the maximum absolute separation across the 23 hinge coordinates at the final time, in radians. The other quantities are signed pulse-minus-control

contrasts in the selected dimensionless activation and tendon-actuator force in newtons. Force follows the model's convention and uses synchronized endpoint forward evaluations, rather than derived values left over from the last integration step.

Table 1: Final contrasts for one simulated right-hand mechanism. All values compare the pulse branch with its matched control at nominal $T = 0.200$ s. The angle value is the largest absolute separation across 23 hinges at that endpoint; activation and force retain the pulse-minus-control sign. Activation and force concern the selected `FDS2_r` actuator. Force uses the synchronized final forward evaluations and the model's sign convention. These are simulated contrasts, not errors against biological measurements. The displayed digits preserve the recorded numerical values, without establishing biological accuracy.[16]

Quantity	Contrast	Unit
Largest absolute final joint-coordinate separation, Δq_{∞}	0.5027347840656727	rad
Selected final activation difference, Δa	0.05205729904534248	none
Selected final tendon-actuator force difference, ΔF	−6.407722136774943	N

Independent auditing of the saved evidence completed 132 standard-library checks without a failed check or model rerun. All required recorded numerical channels were finite and recorded native warning counts were zero. Four stored endpoint mass-matrix spectra were positive. Raw mass matrices and residual operands were not independently exported; their stored diagnostics were checked without independently resolving those matrices. No contact was observed. At the final control endpoint, the recorded soft-limit excess was 0.0004121829780273552 rad, so exact hard-bound enforcement is not established. Neither conservation, numerical convergence nor a physiological response threshold follows.

These contrasts establish a response to the added excitation within this simulated configuration. They are not errors against measured human movement, EMG or force. Two dependent trajectories with 200 time steps supply neither 200 independent experiments nor biological replication. No statistical *p*-value is inferred, and numerical precision in the record does not establish biological accuracy or repeatability across cases.

20.5.3 From a response to a teaching experiment

The run used CPU dynamics. Readback showed a one-thread BLAS pool and dynamics pool size zero; transient compiler parallelism was unmeasured. Its 263 observed loaded-file identities provide bounded identity evidence, not complete system closure. Resource and performance qualification remain unresolved. The reported process high-water RSS of 5,187,464 KiB can retain pre-exec history and cannot be assigned to this model or establish a model-only 3 GiB bound. Tool/native timing near 448 ms and in-probe timing near 491.927 ms have unreconciled boundaries. There is no accepted end-to-end speedup, useful complete latency or full CPU/GPU partition.

The hand's zero sensors leave Myoception—the programme's proposed sensory side of muscle-driven action—without a measurement in this test. Simulated activation, force

and motion do not supply afferent recordings, biological feedback, a learned controller, a trained Large Movement Model or generalization. The four proposed source groups remain neural observations, myography, measured motion and physics-generated examples. Mechanics, controls, calibration, uncertainty, source rights and data lineage require qualification before this generator could enter training, and its contribution would need independent held-out real tests. The amount/diversity hypothesis remains untested.

For Brain2Body, even a later qualified controller would leave the teaching question: what does a trainee retain when trainer commands end? That comparison still needs the original matched information, practice and assistance controls, sham/delayed/unrelated guidance, independent outcomes and a constant compensatory movement bridge where required. This diagnostic contains no trainee, teaching, withdrawal or physical intervention. It selects no training generator and supplies no biological ground truth. LINK's eighteen fits and 720 saved epochs retain both failed admission gates and all five failed benefit claims; protected tests and quarantined sources remain closed.

20.6 Carry the test through teaching and withdrawal

A better movement prior might improve assistance, but the teaching hypothesis requires another comparison. A prospective trainer–trainee study must identify the model's inputs, expert information, feedback, commands, practice and assistance burden. Match those resources against the appropriate simpler controller and original sham, delayed and unrelated-guidance controls. Otherwise an apparent neural benefit could come from supplying the target action, more practice or a less demanding task.

Retention is assessed after trainer commands stop, with any necessary compensatory movement bridge held constant. Immediate performance, delayed retention and unfamiliar task variations remain separate outcomes. A gain that depends on continued guidance may support assistance. A controlled gain that persists would need its own domain and uncertainty; it would still establish neither tissue repair nor arbitrary skill transfer. Simulation, offline replay, measured biological feedback and physical intervention retain distinct evidence and authority.

No scaling result, trained body-wide world model or biological teaching result is reported here. LINK retains eighteen fits and 720 saved epochs, with both model-admission gates and all five benefit claims failed. A broader programme gives those failures somewhere useful to lead: a new, source-qualified comparison that can show exactly which prediction improves, what supplies the improvement and whether a trainee retains anything from it.

21 From a guided action to a retained skill

Catch a ball with outside guidance, then try again after the guidance ends. The second catch asks what changed in the trainee. The comparison is useful only if the task and access to movement remain comparable; removing a necessary movement bridge

along with the trainer would confound the test. This distinction carries the proposed sequence from touch and object handling to NeuroBoxer and NeuroArcher.

The near-term apparatus would record scalp EEG and surface EMG from sleeves on both arms, alongside synchronized, calibrated left and right tripod views. OpenBCI or an equivalent configurable system remains a candidate requiring channel, site, reference, bandwidth, unit, delay and clock qualification. Each event needs its independently observed movement and, where claimed, an appropriate contact or force reference. Surface EMG supplies neither direct force nor afferent nerve activity. A virtual body can predict a response, but its predicted contact or sensory state cannot stand in for those observations.

This left/right hand-task apparatus and the earlier opposing top/bottom dot-map apparatus are different proposed configurations of the motion-capture relay-composite. Their calibration, frames, visibility, occlusion and timing must be qualified separately. Preserve the source of each measured point and the uncertainty of the joined map. The musculoskeletal-simulation physics-engine also needs task- and organism-specific anatomy and mechanics; lifelike rendering cannot establish its function.

21.1 Build coordination through four declared tasks

Touch and movement. Rapid hand rubbing brings contact and movement together. Record the requested knuckle movement and interlaced-hand stretch, with palms outward and arms extended, as separate conditions. Matched observations must distinguish contact-related effects from movement-related effects; the cold-hands metaphor does not establish a cold-exposure result or identify a unique sensation.

Ball transfer and juggling. The proposed hot-potato task passes one juggling ball between the hands, followed by the preferred two-ball juggling task. One limb's action changes the next limb's problem. Independently observed release and catch events, trajectories and timing would test that coordination. Preserve uncertainty when an event or body part is not observed, and compare with cue- and history-based controls.

Punch discrimination and NeuroBoxer. Independently labelled punch classes would test discrimination and movement prediction before becoming the NeuroBoxer virtual game. Boxing includes hand and wrist stabilization as well as arm motion. A prime-era Mike Tyson opponent is a proposed virtual game character. A virtual opponent supplies no physical opponent or evidence of safety. Rights-qualified expert footage can inform movement and tactics, but supplies no expert neural, EMG or force recordings.

Toy archery and NeuroArcher. Toy archery would add explicit grasp, asymmetric two-hand roles, draw under opposing loads, release and return. Its proposed arrows have small, cup-shaped suction tips and are carried in an over-arm pouch with multiple arrows. NeuroArcher would use that coordination to act on a virtual target.

Success still requires a declared outcome and an independent comparison, rather than agreement between simulated components.

The whole snap is a further coordinated showcase, from arm-out/palm-down rest through elbow and forearm swing and turn, wrist and hand upward during the snap, then downward return. The cup reach remains a simpler precision example. The motion, contact and anatomy needed for either task are specific; a human-looking animation cannot demonstrate a monkey's ability to perform the snap.

21.2 Separate motion control, neural contribution and physical writing

The first game milestone uses body motion to control a virtual body. A later EEG-only milestone would ask whether permitted brain observations contribute to the specified action. Withheld EMG and measured movement cannot become predictor inputs through alignment, filtering or cue leakage. Eye activity, covert muscle activity, residual movement and shared task information need their own controls. Keep the original brain-only, body-only and paired comparisons, target-persistence baseline, partition rules and success criteria.

Aim to make the declared intent-to-action mapping as close to one-to-one as possible by measuring its intermediary. State the participant, session, condition, allowed context, tolerated error and independent reference. Reduced ambiguity would be a task-specific result; it would not identify a unique hidden intention or establish the best achievable mapping. Reverse reconstruction of recorded EEG features remains conditional and cannot deliver sensation.

The proposed later physical writer is a separate experiment. EEG and surface EMG record, whereas neuromuscular electrical stimulation (NMES) and functional electrical stimulation (FES) denote stimulation approaches that can produce contractions or functional movement. Sensation delivery needs its own writer or sensory transducer. Device, clinical, consent and physical admission remain separate, and physical spinal scope remains read-only. These writers and actuators are unqualified and outside the existing acquisition/virtual-work budget scope; no recording or intervention is reported here.

21.3 Ask what remains after the help ends

A proposed blindfolded condition changes visual feedback. It cannot independently establish neural contribution, causal control or retained skill. Those claims need matched comparisons, including a trainer-off test that preserves any movement bridge needed to perform the task. Measure immediate performance, delayed retention and transfer to specified unfamiliar conditions separately. Report remaining support, uncertainty and alternative explanations.

The model's training and the trainee's learning also need separate evidence. Qualified paired data could fit predictions; optional control-policy reinforcement learning needs states, actions, transitions and rewards. A simulated reward is not evidence of biological teaching. Final held-out outcomes remain outside fitting, fine-tuning, rewards and model selection; frozen inference is distinct from a separately reviewed

adaptive update. The retained LINK comparison failed both model-admission gates and all five benefit claims. These prospective tasks do not revise that result or establish retained learning, tissue repair or a cure.

22 Build a brain–body system we can test

The proposed architecture becomes a first experiment only when its measurements are specified. Begin with one hand and one movement that can be measured well. Record the brain and muscle signals, capture the movement, and compare the virtual hand's prediction from one declared input side with that independent measurement. The error identifies what the next experiment must resolve.

The apparatus would use non-invasive EEG, surface EMG, a calibrated motion-capture relay-composite, an open musculoskeletal-simulation physics-engine with a lifelike view, and an open multimodal model trained on eligible neural and muscular structure and activity data. The EEG/EMG observation pair and the movement/physics pair would meet inside one conditional learner. Intent-related brain signals and muscle activity would supply observations; measured movement would check what the body actually did. Predictions from either end must answer to observations outside the model. Form constrains what could happen; measured function tests the prediction.

22.1 One movement. Two directions.

The forward test uses EEG and declared task context to predict supported muscle activity and movement on withheld recordings. The reverse test uses muscle activity and movement to predict recorded EEG features within that task. Compare both predictions with independent measurements. Agreement between the models cannot substitute for that test.

Movement measures what the body did. It narrows the explanations of brain signals that remain plausible without uniquely identifying intent or subjective experience. Retain stimulus, task, behaviour and separately reported context. Show uncertainty instead of displaying a generated neural pattern as the observed mind.

Surface EMG records muscle electrical activity. A claimed incoming sensory or feedback pathway needs evidence of its own. The virtual body cannot supply that missing observation. Keep recordings, predictions and simulation identifiable throughout the test.

22.2 A bridge must account for the spine

The spinal cord helps shape the response. Muscle output depends on sensory input, movement context, local circuitry and descending input; some spinal pathways can retain learned changes. Reproducing a specified spinal function would therefore require more than forwarding its signal: a brain–body bridge must account for the processing it aims to reproduce.

A changed response can reflect immediate reflex modulation, lasting spinal plasticity or peripheral adaptation. Those explanations need separate tests. A change during the task does not establish retained memory, and a retained change does not by itself locate that memory. Human reflex conditioning and constrained mouse adaptation supply evidence for different claims. [4, 17, 18] The human studies do not establish brain-independent learning; the mouse preparation does not demonstrate unrestricted walking or reaching. Our human comparisons use abstracts and selected results, discussion or methods passages. Relevant mouse-study passages were also examined in the author manuscript; equivalence to the final published edition and the complete methods and statistics have not been established here.

The periphery adds a different question: did the sensory signal itself change? Passive proprioceptive training supplies a task-specific peripheral comparison; its psychophysical and spindle-afferent experiments were separate. [19] Our passive-training comparison uses abstracts and selected results, discussion or methods passages. None of these studies demonstrates universal skill transfer or complex memory stored in a peripheral axon.

The next comparison must choose a function we can measure. Does the controller respond appropriately to the supported context? Does an adaptation persist through the declared test? Could peripheral or mechanical changes explain the outcome? Keep recordings, estimates and simulated state in distinct visible layers. Surface EMG cannot supply a missing sensory-nerve or spinal channel; an explanatory cutaway cannot register recorded neural sites.

Externalization remains the research aim: can a specified spinal function be reproduced outside its usual pathway? Replacing a computation, stimulating a pathway, controlling a simulated limb and restoring biological function would answer different questions. Each needs its own task- and site-specific evidence; physical spinal scope remains read-only. Anatomical access does not establish neural function, increased excitability does not establish useful plasticity, and no fixed percentage of a complex skill can be allocated to brain, spine or periphery from these proposals.

22.3 Reading a signal and writing an output

The first prototype would test the recording and prediction route with EEG, EMG and independent movement measurements. To deliver feedback or act on a body, it would also need the separate sensory or action route specified in Section 19.9. That method distinguishes indirect sensory feedback from direct neural writing and virtual control from robotic or later biological actuation, with the device, measurement, safety and clinical evidence each would require.

Physical sensory writers, transducers and body actuators remain excluded from this acquisition and virtual-work plan, unqualified and unpriced. Their absence defines what this first prototype could test. The consolidated interface method also states the excluded procedures and permissions: the proposal authorizes no stimulation, participant acquisition, biological actuation, new equipment purchase or training.

22.4 The first prototype has to earn the next one

The lean tier tests whether the selected signals and calibrated movement reference can be synchronized and used for a bounded task. The serious-prototype tier improves acquisition, independent references and source-specific bidirectional evaluation. Expand cohorts, tasks, instrumentation and compute in the research-lab tier only when earlier evidence justifies the next step.

Every tier would include equipment, calibration, accessories, storage, integration, appropriate safety and consent work, independent references and contingency. Planning figures would be stated in USD, with current, dated vendor sources and a dated basis for any currency conversion. Listed prices, engineering allowances and items requiring quotes would remain distinct. Each budget defines a proposed scope; it authorizes no purchase and promises no complete neural reconstruction.

22.5 Let the model earn the claim

Which open recordings can support the next model comparison? Their rights, version, coverage, clock, units and missing modalities determine the answer. Training would proceed only under an independently admitted protocol, with test participants, sessions and tasks kept outside model selection. Simple baselines would give the added model a comparison to beat. Examining artifact leakage, absent inputs, timing errors, physically invalid solutions and uncertainty would keep that comparison answerable to its actual conditions. Earlier failures and reserved recordings remain constraints on it. The new ambition must answer to the evidence already kept.

Begin with constrained virtual hand or foot control. Robotics, remote operation, unfamiliar effectors and extra limbs remain later research directions, each requiring separate physical, latency and safety evidence. A successful movement prediction supports a movement claim. Biological and perceptual claims require the corresponding broader evidence.

Rights-cleared images and lifelike interactive models would make the apparatus and its predictions inspectable. The comparison should identify the observation, the prediction and the error, with the conditions that limit each claim. A convincing image cannot supply a missing measurement or demonstrate biological function. The next experiment must make those distinctions visible enough to test.

23 Testing the Proposal

A prediction becomes testable when the question names an observable outcome and an independent reference can check the implemented answer. The measured error then gives refinement a direction. Apply that reasoning to acquisition, architecture, training, body mechanics, visual models, interfaces and communication. Synthetic augmentation can support the work only with real ground-truth support; it cannot invent missing training evidence or replace independent held-out real measurements for biological claims. Preserve failed trials and the original thresholds so that a more attractive result cannot quietly redefine success.

Interoceptomics, the proposed study joining internal-body sensing with connection maps, may extend the programme to heartbeat, breathing and other internal conditions. It needs distinct measurements and claims; the movement-first rationale does not directly reveal all sensation or private experience. Useful limited interfaces remain worth investigating even if a universal translator never emerges.

Return to the guided hand. A correct tune would establish a performance under specified help; a delayed, controlled trainer-off comparison would ask what the trainee retained. Better measurements and body-grounded tests might connect those two achievements. Current results establish neither arbitrary skill teaching nor complete biological virtualization, and the current development models fail their benefit gates. The next generation must account for what moves, what help causes the improvement, what remains without that help and where the explanation breaks. That is the experiment the ambition still owes us.

24 Definitions and Lexicon

The Scientific Article definition and lexicon supplement is derived from the shared Brain2Body registry identified by version and fingerprint at the start of that supplement. The fingerprint identifies the exact definition source; it is not a scientific score. Preferred names, historical lookup labels, proposed scope and implemented software retain their distinct meanings. The Simple English Edition has its own supplement using the same registry. Necessary specialist terms are also explained where used in this article.

25 References and Evidence Status

Primary scientific sources are linked beside the claims they support. The accompanying bibliography preserves verified earlier identities, including the organoid paper's correction, and adds the ground-truth case-study sources used here. A published experiment supplies different evidence from a dated developer disclosure or an observation of pinned source code; those distinctions govern the claims throughout this article.

The accompanying computational results and limits describe unpublished numerical and software studies. The companion NeuroTranslation manuscript is also an unpublished working draft; neither source is peer-reviewed experimental support. The historical conceptual figure explains a relationship rather than supplying new biological evidence. The edition note records the remaining scientific and publication limits.

Historical Conceptual Framework Illustration

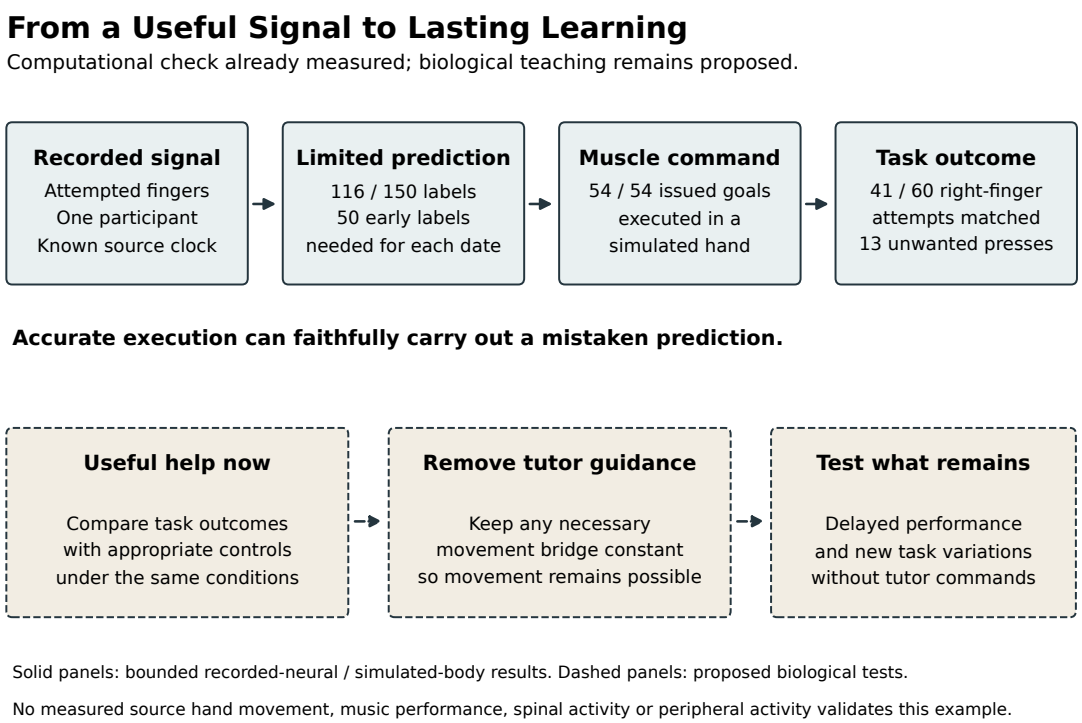


Figure 1: The solid route follows the earlier recorded human finger-attempt computation: 41/60 intended contacts and 54/54 executions of issued commands. Predicting the intended contact and executing an issued command answer different questions. The biological branch remains proposed. This historical schematic supplies no current evidence of anatomical or functional correspondence, lifelike-3D fidelity or biological validity. The separate macaque development study saved 18 runs and 720 epochs; both model gates and all five benefit claims fail. Neither study establishes biological teaching, full-body mapping or retained mastery.

Scope of this source edition

These unrendered manuscripts retain the computational studies and add the expanded training proposal and one recorded numerical hand diagnostic. The diagnostic compares two simulated trajectories. It supplies no new neural fit, reserved-data evaluation, biological teaching or validated whole-body model. The definition supplement and source projection now contain the current registry's 200 entries, revision 34, including their distinct proposed scopes. No PDF or native export has been generated from this source edition.

Whole-manuscript scientific, editorial, citation, rights, font, layout, native and export assessment remains unfinished. Current vocabulary does not change the original numerical results or establish new model capabilities.

Evidence, scope and edition status

A computational failure can narrow the next experiment; it cannot qualify a biological interface. These unpublished working manuscripts report computational studies and set out a separate proposed paired experiment. This edition reports no new fitting, reserved-data evaluation, model promotion or biological teaching. The common optimizer has not been independently certified. The original tolerances and reserved recordings remain unchanged, and each failed gate stays failed. Scientific, clinical, physical, real-reference-series, rights and runtime claims still require their original validation. Independent assessment of the whole manuscript's science, sources, citations, figures, actual rendered outputs and native exports remains unfinished, as do publication checks. This computational failure report establishes no readiness for publication.

This source edition uses the current shared glossary, revision 34, with 200 entries. Its current JSON projection retains every canonical term field, alias, proposed-name record and source history. The definition supplement keeps compatible earlier explanations and updates current role meanings; the six additional entries remain proposed programme terms. The full source fingerprint and current definitions appear at the start of the supplement. Current terminology does not revise earlier measurements, reopen reserved tests or establish a proposed capability.

A generated illustration can explain a mechanism; it cannot replace the working model or measured evidence.

The requested public byline is `@eliminatenoise`. Author identity, applicable affiliation and repository submission requirements must be resolved before publication. No affiliation or publication acceptance is implied.

Brain2Body Definitions and Lookup Index

Canonical registry revision 34; 200 entries; SHA-256 `3fffa79bc9f404caa7d4910e9e37de5a97371fdc1ad11bc54155d309ec5f5394`.

A measurement, a computer state and a claim about biology need different words. These entries explain the current programme meanings. A proposed name does not establish a working interface, a trained model or a biological result.

Trainer and trainee name the current guidance roles. A machine-learning model can still be a learner; its fitted parameters do not measure a person's retained learning. Historical names and original source quotations remain historical.

Lookup labels help locate an entry. Some name related or contrasting concepts, so they are not interchangeable. The current JSON projection preserves every canonical term field, alias, source and definition history.

Physical scale and units, computational detail and statistical aggregation remain separate. Missing ranges or measurements stay unspecified. Both paper editions retain the same definitions and scientific scope; explanatory wording may differ.

A checked message between software parts

A structured message and interface specify what the values mean and how software parts exchange them.

The checks include recording, ordered channels, quantity, timing where supplied, units and body axes. Passing a check does not prove that the caller's claimed physical source is true.

Current registry lookup labels: Translation packet, typed port and API; packet; translation port; typed interface; API; application programming interface

A group of recorded neural features

Several recorded neural features considered together when predicting an outcome.

Their joint pattern can help predict movement. That does not establish that the same pattern has the same meaning in another animal, task or day.

Current registry lookup labels: Neural population; population decoding; population model

A map of physical connections

A structural map records physical connections. A functional recording measures activity over time, such as which recorded features produced events during a task.

The structure can guide a model, but it cannot supply missing time-recorded activity. A drawing, connection map and synchronized functional recording have different evidential roles.

Current registry lookup labels: Structural connectome and anatomical prior; connectome; connectomic connectivity; anatomical prior; structural connectivity; structural reference; recorded function; structural versus functional measurement

A model spanning the body

A future model connecting relevant observations and actions across the brain, spinal cord, nerves, muscles and movement.

The project does not yet have a matched voluntary recording across that whole chain. Separate datasets cannot fill the missing measurements.

Current registry lookup labels: Body-wide functional model; body-wide neural fusion; whole-body neural translator

A specified human-related feature

Studying a defined human-related biological component or feature in a model organism.

This can overlap with adding donor cells. One selected immune or neural feature does not make an animal a miniature human or establish human-like cognition.

Current registry lookup labels: Humanization; humanized; human-related feature

Ability that remains after guidance is removed

A learned ability that remains without the training guidance and can be tested again later.

Remove trainer commands while keeping any bridge needed to express the trainee's own control. Switching off that required route can prevent movement regardless of what the trainee learned.

Current registry lookup labels: Retained learning; retention; independent pupil control; retained skill; pupil-alone; necessary bridge held constant; pupil-alone with necessary bridge constant; independent learner control; learner-alone; learner-alone with necessary bridge constant; independent trainee control; trainee-alone; trainee-alone with necessary bridge constant

Action2World

The hand reaches; the cup moves or stays still. Contact, load and the measured outcome tell us whether the action did what we intended, and that feedback shapes the next attempt.

The relationship between an action and its consequences for the body and surrounding environment, including the sensory information those consequences produce.

A predicted consequence is not an observed outcome. Specify the object, reference frame, task, units and uncertainty. Footage supplies movement examples without supplying the performer's neural, muscle or force measurements.

Organizational level: whole system; network; signal/feature

Current registry lookup labels: action2world; action to world; action-to-world

Activity state inside the muscle model

The model's internal state that responds to a command and affects its force.

This state can change more slowly than the command. The displayed activation comes from the computer model, not recovered physiological activation from the macaque recording.

Current registry lookup labels: Muscle activation; activation; activation dynamics

Adjustment using evidence from the new setting

Changing or checking a model using examples from where it will be used.

Adjusting a decoder, checking probability scores and measuring a body's physical properties are different kinds of calibration. The new-day model uses labeled examples.

Current registry lookup labels: Calibration; recalibration; calibrated transfer; uncalibrated; physiological calibration; physiological fidelity; subject calibration

Agent2Agent

One software agent passes declared information to another.

A proposed link carrying specified information between software agents.

Name the agents, message or representation, direction and tested outcome. Software communication does not by itself supply a biological recording, sensory writer, shared experience or demonstrated skill transfer.

Current registry lookup labels: agent2agent; agent-to-agent; agent to agent

Agent2Brain

An agent would give a person feedback during a stated task. Tests would have to distinguish help during that task from learning that remains afterwards.

A proposed path from a software agent to a person through specified feedback or a separately qualified neural output interface.

Feedback would reach a person through an ordinary sensory route or a separately qualified direct neural interface. Those routes need different devices, evidence and admission: feedback on a screen is not direct neural delivery. EEG and EMG record; they do not write. The proposed agent-to-person path leaves the reverse observation and feedback direction intact. Help during a task must be tested separately from learning retained after it ends. Agent2Brain establishes no stimulation or physical-action authority. Physical spinal scope remains read-only.

Current registry lookup labels: Agent2Brain?; agent2brain; agent2brain?; agent-to-brain; agent to brain

All outputs inside their ranges together

How often every output of interest falls within its stated range in the same case.

A 90% range for each separate output does not guarantee 90% coverage of all outputs at once or of a whole body.

Current registry lookup labels: Simultaneous coverage; all-output coverage; whole-body coverage

An independent way the model can move

One independent model movement, such as rotation at a joint. The formal shorthand is DOF.

The native arm has seven coordinates and no separate finger movements. The human replay controls two goals while bracing other independent joints.

Current registry lookup labels: Degree of freedom; DOF; DOFs; independent coordinate; braced joint

Analysis timing and biological travel time

An analysis offset aligns two recorded quantities. A conduction delay concerns how long a signal travels through biology.

The 40 ms sensory offset estimates an earlier limb state. The motor target's 4 ms offset comes from sampling. Neither measures neural conduction time.

Current registry lookup labels: S1 alignment offset and conduction delay; 40 ms sensory lag; S1 lag; 4 ms sampling offset; conduction delay; physiological latency

Animal drawing and measured anatomy

An illustration explains a proposed animal or body. An empirical anatomical atlas is built from verified observations of anatomy; it cannot be established by a convincing drawing.

The antechinus illustration is a proposal, separate from acquired macaque and human sources. The implementation has no matched validated antechinus neural/body atlas and makes no living-animal learning claim from the image.

Current registry lookup labels: Antechinus proposal; antechinus; marsupial proposal; hypothetical organism; Proposed marsupial setting; anatomy illustration; illustrative model; empirical atlas; anatomical atlas

Another brain area used in the finger study

A brain area recorded alongside the movement area in the human finger study.

The formal name is posterior parietal cortex, or PPC. The study provides attempted-finger labels, not measured trajectories of executed finger movement.

Current registry lookup labels: Posterior parietal cortex; PPC; Parietal brain area

Assumed repair to the native muscle model

A separately named model version replaces unusable zero-rate properties with the software's default values.

The source uses a named Schutte muscle model. Restoring 117 default properties permits numerical tests but does not measure the subject's muscle response rates.

Current registry lookup labels: Schutte muscle properties and default-rate fork; Schutte; zero-rate; assumed-rate fork; class defaults

Attachment order and the chosen route segment

The ordered connection points and the numbered segment to which a wrapping rule applies.

The rule uses the source model's attachment order. The SDK counts from one and includes both range endpoints; the correction does not fit arbitrary points to an error frame.

Current registry lookup labels: Attachment topology and SDK range; attachment topology; one-based inclusive range; humerus-to-ulna segment

Axes, pushes and turns

A coordinate frame states the axes used for movement. A handle wrench combines forces and turning moments measured in that frame.

The handle record has three force and three moment components. An unknown mapping into the computer-body axes prevents treating those forces as verified simulator inputs.

Current registry lookup labels: Coordinate frame and handle wrench; body frame; XY; physical frame; handle wrench; forces and moments

Biological tissue repair

Restoration or replacement of biological tissue or its function, supported by direct biological evidence.

Better simulated tracking, assisted movement and retained behavior are different outcomes. The prototype does not demonstrate neural tissue repair.

Current registry lookup labels: Regeneration; tissue regeneration; repair

Biological virtualization

For a movement model, matching the path is one question; matching the living system's response to an intervention is another.

A claim that a digital model reproduces specified biological behaviour and causal dynamics within an independently tested domain and stated error bounds.

Independent ground truth must establish measured error and repeatability within declared conditions and tolerances. For causal dynamics, the comparison also needs independently measured responses to appropriate causal interventions; a similar-looking trajectory alone cannot establish them. Approximation can make a model plausible and may contribute to a validated model when its assumptions, errors and domain are tested and disclosed. A musculoskeletal simulation is itself a model to test, not independent empirical ground truth. Numerical detail and repeated verifier results guarantee neither biological accuracy nor complete physiology, subjective continuity, consciousness transfer or implemented Brain2Body capability. This biological claim is distinct from operating-system virtualization.

Scope: validation: Specified observed behaviours and causal dynamical responses under declared conditions.

Organizational level: whole system; network; organ; local circuit; cell; signal/feature

Computational granularity: Declared model and brain-body interfaces

Statistical aggregation: Declared validation aggregation; separate from physical scale

Current registry lookup labels: virtualization; virtualisation; biological virtualisation

Body dimensions and resistance to motion

Mass is the amount of matter modeled. Inertia describes resistance to changes in motion; anthropometry means body dimensions.

The scaled model keeps assumed source masses and applies geometry-related inertia updates. Its length fit does not identify the animal's actual masses, strengths or individual bone dimensions.

Current registry lookup labels: Mass, inertia and anthropometry; inertia; mass; anthropometry; subject scaling; specimen scale

Body position stored by the simulator

The model's stored configuration, such as its joint positions.

The formal variable name is qpos. Its entries depend on the model and are not automatically measured joint angles. The matching velocity variable, qvel, is explained separately.

Current registry lookup labels: Generalized position and velocity; qpos; joint state

Body-model software and its programming tools

OpenSim provides tools for body models, geometry, scaling and simulation. Its software development kit, or SDK, lets code work with those models.

Checks on these tools establish the stated model's numerical behavior. They do not certify the exact upstream model version or the recorded animal's anatomy.

Current registry lookup labels: OpenSim and its SDK; OpenSim; SDK; software development kit; native SDK; OpenSim 4.5.2

Body-position feedback

Proprioception means sensing body position and movement. For example, it helps a body monitor where a limb is. Biofeedback uses bodily information to guide action or learning.

The implemented correction uses simulated body position. It does not add measured peripheral nerve signals or show that a biological feedback process taught a retained skill.

Current registry lookup labels: Proprioception and biofeedback; proprioception; simulated proprioception; biofeedback; Body-state sensing and feedback; proprioceptive feedback

Body2Action

The proposed link from body mechanics and sensory interaction to a specified, independently measured task action or outcome.

A task outcome makes this proposed link testable: the body's forces, contacts and timing have to be measured in the relevant context. Body2Action connects body mechanics and sensory interaction to that specified action or outcome; it establishes no working controller. World labels are historical conceptual lookup references, not a strict synonym for every environmental interaction.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Current registry lookup labels: Body to action; Body2World; Body to world

Body2Body

A measured movement can constrain a digital body; a digital body can suggest a movement to test. Body2Body asks which quantities can travel between those settings and whether the resulting movement matches an independent reference.

A proposed research direction relating specified physical-body measurements to digital body models and testing mappings or feedback from those models to declared target bodies.

The user's physical-to-digital and digital-to-physical wording names a proposed direction, not a proven symmetric inverse. Observation-to-model fitting, retargeting and a separately qualified physical output have different requirements. Homologous and mechanically analogous features can support unpaired pretraining or transfer when their mechanics, scale, contacts, timing and missing features are declared. That learned relationship is not an actually measured neural pairing. Keep inferred and simulated quantities distinct from measurements. Biological virtualization still requires independent biological outcomes in a stated domain; no real body actuation, retained skill or whole-organism equivalence follows here.

Current registry lookup labels: Body to body

Brain imaging related to blood oxygen

An imaging method, called fMRI, that measures signals associated with changes in blood oxygenation.

It does not record individual electrical spikes. The paper discusses a published model for this kind of image, not a project result showing that brain scans and spikes are interchangeable.

Current registry lookup labels: Functional magnetic resonance imaging; fMRI

Brain pad

A pad shows the proposed interface at the brain.

The proposed brain-side (cortical) interface, drawn as a pad.

A contextual display label in the user's Interoceptomics visual instructions, not a general clinical synonym. The name establishes no acquired neural data, registration, tissue access, function, safety, clinical use, species or physical dimensions.

Current registry lookup labels: brain pad; cortical prosthesis; cortical interface; cortical interface prosthesis

Brain stimulation

An intervention that applies a specified physical stimulus to alter brain activity.

The method must be named, such as transcranial magnetic/electrical stimulation or intracortical microstimulation. These differ in access, risks and capabilities. EEG records; a scalp EEG headset does not provide general sensory writing or safe skill upload. Broad neuromodulation is not identical to precise task-content encoding.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

Source identifier: flesher2021

DOI: 10.1126/science.abd0380

https://pubmed.ncbi.nlm.nih.gov/34016775/

Research precedent for motor decoding plus sensory feedback through intracortical microstimulation; invasive and not a consumer EEG capability

Current registry lookup labels: Neural stimulation; Neurostimulation

Brain2Agent?

A recording supplies observations that an agent may use for a stated task.

A proposed path from specified brain observations to a software agent through declared recording and interpretation interfaces.

Recorded brain observations would reach an agent through a declared recording and interpretation interface. Where applicable, feedback toward the agent remains distinct from Agent2Brain's proposed agent-to-person path. Neither direction identifies unique private intent or experience. Recording supplies observations; inference and simulation interpret or model them, while actual sensory delivery requires its own interface. Brain2Agent? remains an open proposal.

Current registry lookup labels: Brain2Agent; brain2agent; brain2agent?; brain-to-agent; brain to agent

Brain2Body

Connecting brain and body function; the 2 means to.

The numeral 2 is the preferred spelling. The programme includes body-to-brain feedback as well as outward pathways; the name does not establish complete biological translation or novelty of body–brain research.

Organizational level: whole system; network; local circuit; signal/feature

Current registry lookup labels: brain2body; Brain2body; brain 2 body; brain-to-body; brain to body

Brain2Brain

Information links between two brain or neural systems.

The project's label for studying specified information links between brain or neural systems, including trainer–trainee comparisons.

Brain2Brain is the numeral-style public label for the project's existing brain-to-brain research direction. Each experiment must specify its signal, task and controls; the name itself establishes no shared thoughts, general skill transfer or successful biological teaching.

Organizational level: whole system; network; local circuit; signal/feature

Current registry lookup labels: brain2brain; brain 2 brain; brain-to-brain; brain to brain

Brain2Motion

The proposed movement pathway connecting Brain2Body with Body2Action and the resulting measured motion, including the returning sensory feedback.

The label concerns the movement part of the wider programme, not every sensory/cognitive function. Brain/spinal/peripheral observations, muscle activity, body mechanics and external outcomes stay distinct. A decoded motion signal, simulation or lifelike animation is not a complete physiological explanation or validated skill transfer.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Current registry lookup labels: Brain to motion

Brain2Spine

A movement command meets a spinal cord that already regulates reflexes, coordinates muscles and adapts through experience. Understanding that middle step helps explain how the same intention can produce different actions.

The pathway from brain activity into spinal processing: descending signals influence circuits that combine them with sensory input and the body's current state.

A scalp EEG recording does not identify every descending signal or measure the spinal circuit's hidden state. The physical spine remains read only. This label does not name an admitted stimulation interface.

Organizational level: whole system; network; signal/feature

Current registry lookup labels: brain2spine; brain to spine; brain-to-spine

BrainNet

A network of specified links between brain or neural systems.

A working network metaphor for connected brain or neural systems within the programme.

BrainNet pictures specified links between brain or neural systems as a network. It is a programme metaphor; it establishes no validated universal neural network, uniquely coined field or available trademark.

Organizational level: whole system; network; local circuit; signal/feature

Current registry lookup labels: brainnet; Brain Net

Brain–Body Interface

A brain recording and a measured hand movement answer different questions about the same reach. A Brain–Body Interface would connect the stated observations and any separately qualified outputs so that the proposed relationship can be tested.

A proposed interface linking specified brain observations with body observations or control outputs in a declared task, with each direction and physical route identified.

This names an interface role, rather than the Brain2Body programme or a model spanning the body. A Digital Bridge is the existing selected-information route around an interruption or between specified recording and control channels; it is related, not an automatic synonym for every Brain–Body Interface. Recording, inference, simulated control and physical output require distinct evidence. EEG and EMG record; a feedback or stimulation output needs its own qualified interface. The label establishes neither complete coverage nor a bidirectional biological loop. No new acronym is introduced.

Current registry lookup labels: Brain-Body Interface

Choosing a model and deciding if it passes

Choosing settings and applying a stated pass rule using the validation data.

The frozen-transfer rule requires a prediction score at least 0.01 above every specified control. A failing candidate stays rejected rather than being given an easier success rule.

Current registry lookup labels: Validation selection and admission; validation-selected; acceptance criterion; admission; failure criterion

Command sent to a modeled muscle

The input command that starts the modeled muscle’s activation process.

The command is bounded and nonnegative here. It differs from the resulting activity state, recorded muscle electricity and instructions for a biological intervention.

Current registry lookup labels: Muscle excitation; excitation; actuator command; muscle control

Communication route around an interruption

An interface carrying selected information around a modeled break or between stated recording and control channels.

A bridge can compensate without repairing tissue or supplying muscle energy. The virtual interruption does not establish a biological injury model or treatment.

Current registry lookup labels: Digital bridge; compensatory bridge; brain–spinal bridge; BCI; brain-computer interface; virtual pathway interruption

Comparable roles in different organisms

Comparing roles such as reaching or support while keeping each organism's anatomy explicit.

Related roles do not imply identical neurons or muscles. A combined fictional body cannot replace the measured sources and checks needed for each real organism.

Current registry lookup labels: Organism role and homology; organism adapter; limb role; functional coordinate; homology; biped/quadruped

Comparisons that test whether guidance matters

Matched replay uses the relevant recorded activity under matched task conditions. Stale or unmatched replay uses activity from a different time or condition. No-link and inactive or sham comparisons test whether the proposed information route adds benefit.

These are proposed comparison types, not new animal results. Shuffling disrupts timing; a yoked comparison gives matched help from another case. Guidance amount, timing and later trainee control need their own stated comparisons.

Current registry lookup labels: Shuffled and yoked controls; shuffled-input control; time-shuffled; yoked signal comparison; Matched comparisons with altered information; matched replay; matched own-session activity; stale replay; unmatched replay; no-link; sham; inactive comparison

Completed time interval for counting events

A stated time interval groups events into one count. A half-open interval includes its start and excludes its end.

Inputs use completed intervals. The exact boundary rule, interval duration and timing must remain explicit rather than being guessed from the display.

Current registry lookup labels: Completed time bin and half-open binning; bin; 20 ms bin; 50 ms bin; half-open; bin end; cadence

Computer processors, memory and file size

A CPU is a general processor, a GPU is another kind of parallel processor, and RAM is working memory. Bytes and GiB measure digital data size.

The training described in this source edition used four CPU threads; GPU acceleration was neither verified nor used. AI-assisted coding does not establish independent biological validation or an authorship credential.

Current registry lookup labels: Computing hardware and data size; CPU; GPU; RAM; AMD; AI; Ryzen AI Max+395; GiB; byte; bytes

Correct-label rate and chance comparison

Accuracy is the proportion of labels chosen correctly. A confusion table shows which labels were mistaken for others.

A uniform guess among ten balanced classes has a 10% reference rate. Accuracy and label scores do not prove executed movement or confidence in biological truth.

Current registry lookup labels: Classification accuracy and chance; accuracy; uniform chance; ten-class confusion; confusion matrix

Count-prediction improvement per recorded event

Bits per spike describes how much a count-prediction score improves over a defined baseline, divided by the number of recorded spikes.

The baseline uses the training mean rate. A positive result concerns estimated average counts, not newly recorded activity or verified stimulation instructions.

Current registry lookup labels: Bits per spike; BPS; ΔBPS; bits/spike; likelihood gain

Data for learning, choosing and checking

Separate portions used to fit the model, choose its settings, and check the selected result.

The paper uses local time-ordered splits. If test results have already been inspected, later reuse cannot be described as another independent test.

Current registry lookup labels: Training, validation and test partitions; train/validation/test; train/val/test; split; chronological 70/15/15

Data kept out of a stated step

Data reserved from training or from another specified development step.

Independence depends on what was actually excluded. A previously inspected case does not become a fresh test just because it is called held-out.

Current registry lookup labels: Holdout; held-out; reserved window; independent evaluation

Describing motion and calculating what causes it

Describing a body's position and movement is one task; calculating how modeled forces change that movement is another.

The paper calls these kinematics and dynamics. A replayed pose and a muscle-driven trajectory are different results; neither alone proves physiological accuracy.

Current registry lookup labels: Kinematics and dynamics; kinematics; dynamics; forward dynamics; numerical plant

Ecotype

A population adapted to a coastal habitat is an example of the ecological distinction at issue. Ecotype names the adapted population or form; the surroundings themselves are its environment.

A locally adapted population or form associated with a particular ecological setting.

Ecotype is an established population/ecological concept, not an individual's surroundings or an obligatory third stage after genotype and phenotype. State the organism and evidence for the claimed local differentiation or adaptation. The retained Turesson source is a historical plant example [20], not a complete modern evolutionary theory or evidence about the proposed nervous-system experiments.

Current registry lookup labels: No additional label declared.

Electrical recording in outer brain tissue

Electrodes in the brain's outer layer, called the cortex, record electrical signals outside the cells.

This differs from a scalp recording. It also does not observe every cell in a circuit; the source file defines which events and channels were recorded.

Current registry lookup labels: Intracortical extracellular recording; intracortical; extracellular cortical events; Electrical recording inside the cortex; cortex

Enactome

A reach has a path, a time and a result. An Enactome would organize observations like these so that the action and what followed can be compared across stated conditions.

A proposed organization of specified action observations and the recorded relationships between actions and their outcomes in declared tasks.

Declare which actions, body parts, tasks, units, frames and times the data cover. Keep commands, measured movement, sensory consequences and simulated outcomes separate. An observed association does not establish causation, a complete repertoire or the person's unique intent. This proposed data organization differs from the tentative Enactotype individual-profile label.

Current registry lookup labels: No additional label declared.

EndoHumanisation

Choose a particular human-related feature, then ask what introducing or studying that feature would change in the model organism. EndoHumanisation names that proposed line of work.

A proposed research stage studying specified human-related genetic features within a model organism and testing their relevance to a declared biological question.

The roadmap places XenoTransgenesis at step 1.1. The working interpretation concerns defined genetic features; the actual manipulation and its established scientific name remain to be specified. A selected feature does not make an animal human or establish human cognition, disease fidelity or an appropriate intervention age. Biological relevance, function and animal/participant, ethical, safety and rights requirements need their own evidence and admission. This entry supplies no experimental procedure.

Current registry lookup labels: No additional label declared.

Error range based on validation examples

A range placed around a prediction using the errors seen on the data used for model selection.

The ranges have constant width for each output. They do not guarantee future coverage when examples overlap in time or their errors are related.

Current registry lookup labels: Residual band; empirical residual bounds; validation residual quantile; 90% per-output residual band

Events grouped as one neural feature

The source recording process groups events into features that the model can count separately.

A group is not a verified lifelong identity for one neuron. It must stay separate from unassigned events and from the channel labels used in other recordings.

Current registry lookup labels: Sorted neural unit; sorted unit; sorted cell; sorted feature

ExoHumanisation

Added tissue can survive without doing the job expected of it. ExoHumanisation names a proposed study of a specified human-related graft: what survives, what connects, and what function can actually be measured?

A proposed research stage studying specified human-related donor cells or tissue in a model organism and testing their survival, integration and function.

The roadmap associates step 2.1 with XenoTransplantation. A cross-species graft retains that established meaning; cell survival, anatomical connection, activity and useful functional integration are distinct outcomes. A graft does not establish a human nervous system, a shared mind or whole-organism human equivalence. Species, tissue identity, biological purpose, ethics, safety and rights remain explicit dependencies. This label provides no transplantation procedure or intervention-age window.

Current registry lookup labels: No additional label declared.

Expected event count and its fit score

A model estimates the average event count for given inputs, then a score checks how that estimate fits the recorded counts.

The paper uses Poisson negative log likelihood for that check. A predicted average may be fractional; a good count-fit score does not establish a complete biological spike model.

Current registry lookup labels: Conditional count mean and Poisson likelihood; expected count mean; conditional count; Poisson; negative log likelihood; NLL

External movement model

The external part of the body model, constrained by independently measured surface movement and pose in a declared task and coordinate frame.

The motion-capture relay-composite retains the source views, clocks, calibration and coordinate frame for the declared task. The earlier upper/lower hand concept and the later synchronized left/right tripod arm proposal are different arrangements; each needs its own visibility, synchronization and error checks. The composite is derived. Hidden internal states, forces and intent are not uniquely measured. LiDAR and optical infrared tracking require distinct hardware, calibration and uncertainty qualification.

Current registry lookup labels: No additional label declared.

Extra correction and its scoring rule

A model can add a correction to an existing controller. A reward scores the result; command-change rate describes how quickly the input changes.

The formal names include residual policy, reinforcement learning and actuator slew. A simulator can score tracking or task completion without proving biological learning or correct hidden neural activity.

Current registry lookup labels: Residual policy, reward and actuator slew; residual policy; reinforcement learning; RL; verifiable reward; actuator slew; task completion

File fingerprints

A checksum or hash is calculated from file bytes to detect changes or match a stated reference.

The named methods here include MD5 and SHA-256. A match checks bytes against that reference; it does not prove scientific truth or verify an entire file when only a range was acquired.

Current registry lookup labels: Checksums and hashes; checksum; hash; MD5; SHA-256; source checksum; archive hash

Force in the modeled muscle and tendon

The force produced by the computer model for a muscle-tendon unit, or a force measured by a stated physical method.

The native test reads forces from the running simulator rather than an old lookup table. Those values are still simulated rather than isolated forces recorded in the source animal.

Current registry lookup labels: Tendon force; muscle force; isolated muscle force

Form → function → action

The programme narrative follows structural constraints, the context-dependent processes they support, and independently measured actions or outcomes.

The three words organize the explanation; every molecular and neural level has its own structure and dynamics. Form informs and constrains function but does not determine the full process alone. Feedback returns outcomes to the system; this is not a one-way causal chain or a guaranteed one-to-one mapping.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Current registry lookup labels: Form to function to action

Functional electrical stimulation

Electrical stimulation coordinated to produce a specified functional movement or task.

FES is the appropriate functional-actuation label for the proposed later assisted-body study. It requires separately qualified devices, control/safety limits and specialist-supervised research approval. This registry contains no stimulation settings or self-experiment instructions.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

https://www.nature.com/articles/nature17435

Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Current registry lookup labels: FES

Further training for the target task

Continuing model training so it predicts the desired target, using the relevant inputs and labels.

The paper calls this fine-tuning. Changing software weights is not evidence that an animal or person learned a biological skill.

Current registry lookup labels: Fine-tuning; paired-output fine-tuning

Future testing and analysis of existing data

A future test checks later unseen outcomes under a fixed plan; an existing-data test analyzes outcomes already recorded.

The second source file was chosen before inspection, but its recording day is earlier than the original model's day. Choosing a file in advance does not make it future biological deployment.

Current registry lookup labels: Prospective and retrospective evaluation; prospective; retrospective; prospective biological deployment

Gap between data portions

Leaving out an interval around a split boundary so overlapping windows do not cross into another portion.

The continuous pipeline uses a one-second gap and checks the input and target support. This protects the software split, not every assumption about the original measurement.

Current registry lookup labels: Partition embargo; embargo; boundary embargo

Genotype

A genetic description can record which versions of a DNA sequence an organism carries. Genotype names that genetic make-up, whether the description covers a selected part of the DNA or a wider set. The scope must be stated; the genetic description alone does not tell us every observable trait.

The genetic constitution being described for an organism, with the genetic scope of the description stated.

Genotype is an established scientific term. Its description is not a complete prediction of the organism's observed traits; phenotype also depends on environment and development. The three labels in the user's organism-level example are not three obligatory sequential stages or three fixed individual types.

Current registry lookup labels: No additional label declared.

Hallucination

A model can predict a movement correctly while inventing the biological mechanism it claims produced it.

Invented or ungrounded model content presented as if it corresponded to reality. In programme claims, this includes fabricated biological structure, dynamics or ability presented as established biological fidelity.

Research on AlphaFold3 uses the term for generative-model errors; this programme also applies it to unsupported biological attributions. A disclosed approximation, hypothesis, counterfactual or unvalidated candidate is a different kind of statement, and missing evidence alone does not prove it false. Numerical error concerns the calculation; objective misspecification concerns what is being optimized; causal attribution error concerns the mechanism being credited. Those differences matter when deciding what a result supports. An ungrounded generated answer can happen to be correct without validating its claimed biological mechanism or reliability elsewhere. Informed predictions therefore still require independent performance evaluation. The rolling-dice analogy describes reliance on unverified claims, not literal randomness or equal odds. Calling every approximation a hallucination, or every connectome or simulation fictitious, would erase the distinction.

Scope: usage: Model-generated biological content and claims of biological fidelity.

Computational granularity: Model output and its interpretation

Current registry lookup labels: hallucination; model hallucination

Help during current performance

Help from a controller, replay, trainer or interface that improves performance while it is present.

For example, a trainee can move successfully while following trainer commands. This does not show independent learning or tissue regeneration.

Current registry lookup labels: Assistance; compensation; assisted performance

How a muscle’s pull turns a joint

To relate a muscle’s pulling force to joint turning, the model uses its geometry. Consistent work calculations also relate those forces to small coordinate changes.

The formal terms are moment arm and virtual work. Their numerical checks do not measure biological muscle strength; the human controller uses them to account for dependent shoulder movements.

Current registry lookup labels: Moment arm and virtual work; moment arm; virtual-work projection; dependent shoulder-girdle load

How many labeled examples were supplied

The labeled examples given to train or choose a model, including examples from prior recordings.

The first 5% counts new training labels only. Validation labels, earlier-source labels and extra neural samples are additional information.

Current registry lookup labels: Label budget and supervision; label budget; paired supervision; adaptation labels; validation labels; source-task supervision

How scores from several outputs are combined

Scores can be averaged across individual outputs or first combined within defined groups. In this entry, macro describes the averaging and weighting rule, not the size of a body part, the scope of a nervous system, or the level of detail in a computer model.

The sensory result gives equal weight to five physical output groups. This avoids letting a larger group dominate but does not make their different physical units interchangeable.

Current registry lookup labels: Per-output and group-macro scoring; per-output; group-macro; macro R^2

How uncertain the result remains

The remaining doubt about a prediction, including why it could be wrong or change.

Noise in the instrument, errors in the model, changing conditions and missing observations are different causes. The current ranges do not separate all of them.

Current registry lookup labels: Prediction uncertainty; uncertainty; model uncertainty; measurement noise; drift

Human arm model used for playback

MyoArm is the human arm model used for the displayed muscle-driven replay.

The replay uses 63 modeled muscles and two mapped joint goals. Those quantities are simulated, and the macaque velocity mapping does not prove neural or skill transfer into a human body.

Current registry lookup labels: MyoArm; human MyoArm; MyoArm right

Internal musculoskeletal model

The internal anatomy part of the body model: specified bones, joints and soft tissues, assembled from appropriate registered anatomical measurements.

Proposed complementary X-ray/CT and MRI inform different structures. Anatomy does not directly measure activation, force, material properties or neural function. Wētā FX provides visual/simulation inspiration, not a verified identical acquisition pipeline.

Current registry lookup labels: No additional label declared.

Interoception

Information about the body's internal condition, such as signals related to heartbeat or breathing.

Scope: Sensing the body's internal condition differs from sensing a limb's position and movement. Internal sensing remains a research objective; this prototype does not provide a fully measured body-wide internal signal.

Current registry lookup labels: Sensing the body from inside; interoception; interoceptive signal; internal sensations

Interoceptome

A heartbeat signal and a breathing signal describe different parts of the body's internal condition. An Interoceptome would organize the selected observations and their relationships, retaining how, where and when each was recorded.

A proposed organization of specified observations of internal-body sensing and the relationships being studied among them.

This is a data-organization proposal within Interoceptomics. Its coverage must name the organism, body systems, measurements, timing and task. Measured signals, calculated quantities and model estimates remain distinguishable. Touch and limb-position sensing are not relabelled as interoception, and the suffix does not establish a complete internal-body map or an individual biological type.

Current registry lookup labels: No additional label declared.

Interoceptomics

How does the body sense a heartbeat or its breathing, and how are the nerve connections involved arranged? Interoceptomics brings those questions together in a proposed study, relating observations of internal-body sensing to maps of the connections carrying or processing that information.

A proposed field combining interoception and connectomics to study internal-body sensing and its relationships with neural connectivity across scales.

The programme uses Interoceptomics as an umbrella for this proposed research. Giving the work a name does not establish a validated comprehensive map or a recognized external standard.

Scope: macro: Whole-body, organ and network relationships relevant to internal-body sensing; micro: Local signals, cells, circuits and interfaces relevant to those relationships.

Organizational level: whole system; network; organ; local circuit; cell; signal/feature

Current registry lookup labels: interoceptomics

Keeping each source separate

Keeping track of which recording, channel, time and physical setting a value belongs to.

A channel numbered 12 in one day is not automatically the same cell as channel 12 on another day. A software label also cannot prove where supplied data physically came from.

Current registry lookup labels: Source conditioning; source-conditioned; source-specific; recording-specific

Keeping each trial together

Assigning an entire trial to one data portion instead of dividing it among portions.

This prevents pieces of the same trial from being shared between the relevant splits. It does not remove prior inspection or other source limitations.

Current registry lookup labels: Whole-trial separation; whole-trial-separated; successful trial; trial partition

Keeping up now and computing for later

Real-time computation keeps pace with incoming data; offline computation can take longer and be played back later.

Native planning takes about 4.52 seconds for a 20 ms simulated step and is offline. Smooth cached playback does not prove live planning or biological control.

Current registry lookup labels: Real-time and offline execution; real-time; offline; paced playback; cached reference

Kind of measurement

What a method records, such as electrical spikes, scalp voltage, muscle electrical activity or a brain-scan signal.

These signals measure different things. They cannot be treated as interchangeable numbers merely because all of them relate to the nervous system.

Current registry lookup labels: Signal modality; modality; observation process

Large Movement Model

The shorthand “NeuroTranslate with MoCap MyoSim = Large Movement Model” names the proposed combination of NeuroTranslate, MoCap ReComp and MyoSim Replay. It brings a learned mapping, measured movement and virtual mechanics into one architecture. The equals sign is not a mathematical identity or evidence of a trained model. EEG, EMG, measured movement, virtual mechanics and the trainee retain their distinct roles.

The proposed movement-focused learned model that brings neural structure and functional observations together with internal musculoskeletal anatomy and independently measured external movement in a declared task.

The programme model-family name, analogous in ambition to large language model. It does not replace accurate technical architecture descriptions or mean that every DNN is a movement model. Scans and capture alone do not train a model. Parameter count, scale, generalization and nervous-system fidelity require evidence. System 3 thinking remains the partnership framing. In the October 19 shorthand, MoCap MyoSim combines MoCap ReComp and MyoSim Replay with NeuroTranslate. The equals sign names the proposed architecture; it is not a mathematical identity or evidence that a trained model already exists. These three named groups do not remove the distinct EEG, EMG, measured-movement, virtual-mechanics and trainee roles.

Current registry lookup labels: large movement models; LMM; Large movement model

Lasting change in the nervous system

Changes in the nervous system that can contribute to learning or adaptation.

Updating software weights or controller gains is a different event. The practice interface in the computer model does not measure synaptic change in a living organism.

Current registry lookup labels: Biological plasticity; synaptic plasticity; biological learning

Learning before the final task

Training first on another objective, such as predicting a later neural count, before training for a paired movement target.

The paper calls this pretraining. Earlier source-task labels remain part of the information the model received, even when only new-setting labels are counted in a small budget.

Current registry lookup labels: Pretraining; neural next-bin pretraining

Mapping movement onto another body model

Turn a specified movement quantity, such as fingertip velocity components, into movement goals for a particular computer body. The mapping depends on that body and the chosen task.

Mapping specified movement quantities into goals for a particular body model, with the source quantities, target coordinates and transformation declared. This does not by itself reconstruct full posture or demonstrate biological transfer.

The retained source-specific adapter takes two macaque fingertip velocity components as its source and produces human-model shoulder and elbow goals. Their signs carry direction; scalar speed would discard it. These are goals for that model and task, so the mapping establishes neither cross-species skill transfer nor verified biological function.

Current registry lookup labels: Retargeting; body adapter; organism-specific body adapter; joint goal

Meme-jacking

Borrow the recognition; earn the explanation. A familiar image should help the audience see the scientific point before the script develops it.

Use a recognizable meme or cultural reference to make a scientific point immediately legible, drawing on familiarity already earned by others.

Memification hijacking is the user's expansion. References may come from any online culture, not only films. Apply one purposeful reference per teaching concept without crowding formal methods or labels. Recognition does not grant reuse rights, endorsement, proven comprehension or scientific validity. Preserve the original Meme-ification history.

Meme-ification/memeification are spelling forms; mimification is retained only as the corrected historical transcription.

Current registry lookup labels: Meme-ification; meme-ification; memeification; mimification; meme-jacking; meme jacking; memification hijacking

Mind-Meld

What information could one system give another, and would it help with the task? Mind-Meld names the proposed first Brain2Brain encounter, whose signal and effects would have to be measured.

A proposed first Brain2Brain study of a specified information link between neural systems during a declared task.

The roadmap places this at step 3.1 before later NeuroTransfer studies. The name does not describe fused minds, shared private experience, arbitrary thought reading or demonstrated skill transfer. Specify the source, recipient, recording and feedback interfaces, task, controls and independent outcome. Assistance during a link and learning retained without trainer information need separate tests, with any necessary compensatory bridge held constant. A reported natural connection is a motivating case, not evidence that this engineered link exists.

Current registry lookup labels: No additional label declared.

Mind-Merge

An agent could guide practice by giving feedback about a stated task. Mind-Merge names the proposed study of that interaction: what help does the person receive, and what can they do later without it?

A proposed Agent2Brain study of task-specific feedback from a software agent to a person and its measurable effects.

The roadmap places this at step 4.1. Ordinary sensory feedback and direct neural delivery remain different routes with different interfaces, evidence and admission. EEG and EMG record; neither supplies a neural writer. No merged identity, shared mind, retained skill or physical intervention follows from the name. Preserve the reverse observation path, matched practice and delayed/new-task retention tests, removing trainer information while keeping a necessary bridge constant. Physical spinal work remains read-only.

Current registry lookup labels: No additional label declared.

Mixed-source animal and added tissue

A chimera has biological contributions from different sources. A graft is added tissue; when it crosses species, it is a xenotransplant. An organoid is an organized tissue model.

Survival, physical connection, activity and useful function are separate questions. The proposed comparisons do not establish a humanized whole brain or an implemented treatment.

Current registry lookup labels: Chimera, graft and organoid; chimera; chimerism; graft; organoid; host–graft; xenotransplant; xenograft; XenoTransplantation

MoCap ReComp

MoCap ReComp is Motion-Capture Relay-Composite: the proposed measurement component that brings views of the same movement into a common physical frame. It should preserve which view supplied each point, so the joined map can be compared with its original observations.

The proposal starts with two dot maps of the same moving body: one from above, one from below. Align their time and position in a common physical coordinate frame, then join them into one motion map while retaining the source of each point.

A proposed method that creates two dot maps (3D point clouds) of the same moving body, one captured from above and one from below, aligns them in time and a common physical coordinate frame, then stitches them into one composite motion map while retaining each point's source.

To compare the same movement from above and below, the proposal must align both streams in time and a common physical coordinate frame while retaining each observation's source. Bidirectional denotes those opposing measurements, not neural communication or causal direction. LiDAR ranging and infrared optical motion capture are proposed alternatives; marker triangulation and LiDAR point clouds are different measurements, so the actual modality and camera/ranging geometry must be qualified. The maze geometry calls for measurements perpendicular to the floor, with viewing paths from both sides through material intended to be clear. Transmission and material effects need testing; concentrated and distributed arrays remain alternatives. Hardware, coverage, timing, calibration, spatial alignment, occlusion, motion error and latency require validation against real references. The joined map is a derived record, not a direct observation of hidden joints, tissue mechanics, muscle commands or neural function. Neither successful implementation, complete body reconstruction nor material transparency has been established by the name. The single vertical camera illustration remains schematic.

Geometry: Declare the top/bottom sensor positions, measured object, coordinate frame, viewing/illumination paths and coverage. Preserve the requested perpendicular-to-maze-floor geometry; document deviations rather than inferring them from an illustration.

Modality: Declare LiDAR ranging or infrared optical motion capture, including marker-based/markerless method, camera count, triangulation geometry, material path and actual calibration. No complete reconstruction or physiological writing follows from opposing views.

Timing: Record source timestamps, clocks, acquisition cadence, synchronization error and any interpolation, offline processing or end-to-end latency. No real-time capability is assumed.

Calibration and registration: Preserve calibration data and transformations placing both streams into a common physical frame; measure registration error and keep original observations and reconstruction steps traceable.

Limits: Record overlap, occlusion, missing observations, material effects, uncertainty and failure cases. Opposing measurements alone do not prove complete 3D pose, forces or biological function.

Physical scale: Declare measured distance ranges and units for an actual acquisition; no supported numeric range or physical unit assignment is presently available for this proposal.

Statistical aggregation: Declare any chosen evaluation grouping/averaging rule separately from physical scale and computational granularity.

References and acceptance: Use the Reference Comparison Protocol and unfilled record to identify the real-reference series needed for each acquisition component and the assembled map. Its provenance, rights, organism-stage, physical-unit, modality, clock, calibration, occlusion, contact, independent held-out, error, uncertainty and acceptance fields remain unfilled and HOLD. It supplies no measured series or apparatus validation. The unchanged programme reference criteria and every stronger original requirement govern their respective claims; no acceptance threshold is introduced or relaxed.

Outputs: Retain the top dot map, bottom dot map and joined map as separate traceable outputs. Their timestamps and common-frame transformations must be explicit; do not fill unobserved regions as though they were measured.

Scope: proposed application: Complementary top/bottom motion measurements of a body moving through the maze; concentrated coordinated platforms and distributed arrays remain alternatives; measurement boundary: Observed distances or geometry remain distinct from registered/stitched composites and any model-estimated pose or musculoskeletal quantity.

Organizational level: whole system; signal/feature

Computational granularity: top dot map (3D point cloud); bottom dot map (3D point cloud); time alignment and spatial registration; joined composite motion sequence

Historical spellings name this same proposed capture method. Relay here means passing captured measurements onward; the separate agent-coordination Relay protocol does not describe motion capture.

Current registry lookup labels: motion capture composite relay; motion-capture composite relay; motion capture composite; bidirectional composite motion capturing using LiDAR; Motion capture composite relay; motion capture relay composite; motion-capture relay composite; bidirectional LiDAR; opposed-view infrared motion

capture; bidirectional infrared; motion-capture relay-composite; Motion-Capture Relay-Composite

Model that chooses a label

A statistical model that assigns scores to categories, such as the ten attempted-finger labels.

The selected human model uses logistic classification, but its scores do not establish validated confidence in biological truth. Its reported accuracy comes from development data that have already been seen.

Current registry lookup labels: Logistic classifier; logistic regression; classifier; classification; logistic

Model weights kept fixed

Keeping the fitted numbers in a model unchanged during a defined comparison.

The tested transfer trains a connector while retaining the older main model. Fixed weights make the comparison clear; they do not prove biological sameness across days.

Current registry lookup labels: Frozen parameters; frozen trunk; frozen readout; frozen checkpoint

Modeled route from one muscle attachment to another

The route and length assigned to a muscle and tendon as the model's joints move.

The native arm retains 39 named routes. Their modeled lengths differ from direct measurements of living tissue, muscle commands and force.

Current registry lookup labels: Muscle-tendon path; musculotendon length; modeled muscle length; geometric path

Models that convert inputs or predict outputs

One model can turn an input into a representation; another can use available inputs to predict a named result.

The paper calls these encoders and decoders. Predicting counts from mechanics or mechanics from counts does not by itself show the biological cause of either.

Current registry lookup labels: Encoder and decoder; encoding; decoding; readout; decode; encoder; decoder

Movement brain area

A brain area involved in movement. The paper uses the names motor cortex, MC and M1 for relevant source recordings.

The signals help predict recorded movement-related targets. Knowing the brain area does not identify matching cells across recordings or reveal a whole-body command pathway.

Current registry lookup labels: Motor cortex; M1; MC; primary motor cortex

Muscle electrical recording

Electrical activity recorded from muscles; the method is called EMG.

Muscle electrical activity differs from force and from the computer model's command and activity states. The trained targets here are rescaled signal summaries.

Current registry lookup labels: Electromyography; EMG

Myoception

Sensing muscle-driven movement and the feedback that comes with it.

The programme's term for the sensory side of musculoskeletal contraction and movement, including proprioception, tactile feedback and other task-relevant sensations linked to action.

This is the user's proposed programme scope, not a claim that every relevant signal has been recorded or a new external scientific standard established. Muscle electrical activity, proprioceptive or tactile measurements, neural recordings and simulated feedback remain distinct observations. EMG does not directly record sensory afferents or the full sensory consequences of contraction. Declare each source's organism, anatomy, modality, task, units, timing, uncertainty and measured, derived or simulated status. The wider neural-data ambition includes brain, spinal and peripheral activity and relevant visceral signals; the current focus is the motor–sensory system. No body-wide coverage, unique hidden neural state or subjective experience is established by this term.

Current registry lookup labels: myoception; M-Y-O-ception

MyoSim Replay

MyoSim Replay is the proposed MusculoSkeletal Simulation Physics-Engine. Declared anatomy, mechanical parameters and control inputs would produce movement predictions to compare with independent measurements. The external MyoSim body-model collection is a separate resource. Simulation replay is not a measured biological replay.

The proposed digital model would calculate how muscles and the skeleton move and exert forces, with real measurements used to check its predictions.

The proposed physics-based simulation component that models specified muscles, bones, joints and their mechanical interactions, using declared anatomy, parameters, forces and contact rules, with predictions tested against independent movement and other applicable physical measurements.

In the proposed five-component arrangement, declared anatomy, muscle/control inputs and physical context would drive a simulation of movement and forces. The motion-capture relay-composite would provide traceable measured movement to compare with those predictions. That comparison needs matching timing, coordinates, units and calibration, and tests that preserve anatomical and physical constraints, uncertainty, held-out cases, baseline controls, a series of real references and independent acceptance. Movement geometry alone does not measure muscle activation, internal forces, hidden neural state or subjective experience; force, activation and causal claims need additional independent references. Visual effects (VFX) rendering shows the model but cannot validate its mechanics. This remains a proposed simulation component, not a newly demonstrated implementation or external scientific field. Approximate simulation and biological virtualization verified within declared error bounds remain distinct; this component establishes neither universal precision nor whole-body completeness.

Scope: programme role: Virtual-body mechanics component; initially constrained hand and foot tasks; inputs and outputs: Declared anatomical parameters, muscle/control inputs and physical context produce simulated movement and mechanical quantities. Independently acquired motion and other necessary observables remain references, not simulator ground truth.

Organizational level: whole system; organ

Computational granularity: musculoskeletal model; mechanical solver; measurement-to-prediction comparison

Current registry lookup labels: musculoskeletal simulation physics engine; musculoskeletal-simulation physics engine; musculoskeletal simulation;

musculoskeletal physics model; musculoskeletal-simulation physics-engine; MusculoSkeletal Simulation Physics-Engine

Names used for source subjects

Indy, Han and Monkey L label macaque sources; N1 labels the human finger source.

These labels keep different subjects separate. Similar brain-region or task names do not make them one synchronized recording.

Current registry lookup labels: Source subject labels; Indy; Han; Monkey L; N1

Nerves bringing body information in

Afferent paths carry body-related information toward the brain and spinal cord, together called the central nervous system.

For example, a limb-related sensory signal can travel inward. The source-specific cortical predictions do not measure this whole path or make missing nerve activity observed.

Current registry lookup labels: afferent; afferent pathway; body-to-CNS; CNS; central nervous system

Nerves carrying commands out

Efferent paths carry outputs from the brain and spinal cord toward the body, including muscles and organs.

A prediction of movement is not a recording of that outward pathway. The prototype does not supply a synchronized voluntary sequence through spinal cord, nerves and muscles.

Current registry lookup labels: efferent; efferent pathway; CNS-to-body

Neural decoding

Estimating declared information such as intended action from recorded neural observations using a specified model.

Decode accuracy, channels, source identity, task and held-out domain must be demonstrated. Brain-only control requires exclusion of covert muscle, eye and motion signals; decoding does not write signals into the brain.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

https://www.nature.com/articles/nature17435

Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

Source identifier: flesher2021

DOI: 10.1126/science.abd0380

https://pubmed.ncbi.nlm.nih.gov/34016775/

Research precedent for motor decoding plus sensory feedback through intracortical microstimulation; invasive and not a consumer EEG capability

Current registry lookup labels: No additional label declared.

Neural encoding

Mapping declared information to a representation in neural activity; in a sensory-interface design, constructing candidate stimulation patterns for specified feedback.

A computational encoder is distinct from a qualified stimulation device and evidence that a participant receives the intended sensation. The term does not mean arbitrary thought writing or skill upload.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: A brain-computer interface that evokes tactile sensations improves robotic arm control

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Research precedent for motor decoding plus sensory feedback through intracortical microstimulation; invasive and not a consumer EEG capability

Current registry lookup labels: No additional label declared.

Neural Replay

Revisiting a stored sequence of neural samples with its stated timing and channel order.

This playback is not a live recording or instructions for stimulating a nervous system. A displayed arm movement does not show what the source animal's muscles actually did.

Current registry lookup labels: Neural replay; recorded replay; archived replay; cached playback; Playing back a recording

NeuroAgent Trainer

A software agent would guide a trainee during a stated task.

A proposed software-agent trainer role in a declared Agent2Brain NeuroTransfer study.

Identify the actual software model and feedback route. Computed activations/events do not make a biological trainee digital. Illustrative connections do not establish direct neural writing, successful transfer or retained learning. NeuroAgent Tutor is a historical label; NeuroAgent Trainer is the current name. It describes a proposed software-agent trainer role, without selecting the agent or qualifying its feedback interface.

Organizational level: whole system; signal/feature

Current registry lookup labels: NeuroAgent Tutor

NeuroArcher

A proposed archery game that studies coordinated use of both hands and arms through physical motion control, brain-signal control of a digital body, and a separately admitted later assisted-body experiment.

The near-term prop is a toy bow with small suction-cup tips and an over-arm arrow pouch. Target/creature choice is unresolved. Simulated performance does not establish muscle stimulation or retained biological archery skill.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Current registry lookup labels: Neuro Archer

NeuroBoxer

A proposed boxing game that compares arm-motion control, brain-signal control of a digital body, and a separately admitted later assisted-body experiment.

Prime-era Mike Tyson is the requested virtual opponent and meme reference. A game, recorded prediction, stimulation-assisted motion and retained biological skill learning need different evidence. No actual opponent, head contact, finished game or stimulation capability is claimed.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Current registry lookup labels: Neuro Boxer

NeuroConnectome

A map of specified neural connections used to constrain a model of the nervous system.

Connectome is the established term. The user's NeuroConnectome label does not establish complete coverage or identify functional dynamics, state, plasticity, sensory context or unique thought from wiring alone. Declare species, anatomical coverage, resolution and provenance.

Current registry lookup labels: Neuroconnectome

NeuroHandler

How well can a hand carry out the chosen task, and what information helps it do so? NeuroHandler is the proposed manual-dexterity experiment, with the hand's measured action providing the result to explain.

A proposed manual-dexterity experiment studying specified neural/body observations, task control and independently measured outcomes.

The earlier proposed manual tasks remain: rapid hand rubbing, the requested knuckle gesture and the interlaced-hand stretch. NeuroHandler names the manual-dexterity experiment; it does not establish a finished apparatus or successful model. Define the action, available inputs, physical or simulated setting, comparison and outcome before evaluation. Keep motion control, neural prediction, assistance and retained learning

separate; any physical output requires separate qualification and admission. No dexterity gain, clinical benefit or intervention procedure is established.

Current registry lookup labels: No additional label declared.

NeuroJuggler

Moving each limb successfully does not settle whether the limbs work together. NeuroJuggler is the proposed coordination experiment: define the joint task, then measure the relationship between the limb actions and the result.

A proposed multi-limb coordination experiment studying how specified limb actions are coordinated in a declared task.

The earlier coordination proposal remains a hot-potato transfer between the hands, followed preferably by two-ball juggling. Those are proposed tasks; the final apparatus and qualified measurement setup remain unestablished. The label supplies no successful performance or implemented neural interface. Declare the task, limbs, inputs, timing, body model or physical setting and independent outcomes. Separate coordination from isolated-limb performance, simulated success from biological function, and current assistance from retained learning. Physical output, safety, rights and held-out tests keep their own admission requirements.

Current registry lookup labels: No additional label declared.

Neuromuscular electrical stimulation

Electrical stimulation intended to elicit muscle contraction through the relevant neuromuscular pathways.

NMES is a writing/actuation intervention, whereas EMG records muscle activity. EMG electrodes or an OpenBCI recording board do not by themselves provide qualified stimulation. Neither contraction nor imposed movement demonstrates retained skill.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: Restoring cortical control of functional movement in a human with quadriplegia

Source identifier: bouton2016

DOI: 10.1038/nature17435

https://www.nature.com/articles/nature17435

Research precedent linking implanted cortical recording to a custom neuromuscular stimulation sleeve; not an EEG/OpenBCI or elite-skill result

Current registry lookup labels: NMES; Electrical muscle stimulation; EMS

NeuroReplay

What changes when a trainee receives replay-based guidance? NeuroReplay names that proposed study: compare performance while guidance is available, then test what remains when the guidance ends.

A proposed Stage 0 research direction for studying replay-guided sensorimotor feedback and its effects on current performance and later learning.

This is the new roadmap label for steps 0.2–0.5, with a derived working scope; it does not rename Neural Replay, the existing stored-sample playback entry playing-back-a-recording. A stored sequence can supply a comparison, but playback alone establishes neither delivery to a nervous system nor biological learning. Fix the task, signal, feedback route, timing and equal-practice comparisons; distinguish matched, stale or unmatched replay, no-link and inactive or sham controls. Test delayed retention and new tasks after removing trainer information while keeping any necessary compensatory bridge constant. The questioned antechinus age ranges are not established developmental or intervention windows. No physical writer, procedure or present result is supplied by this label.

Current registry lookup labels: No additional label declared.

Neuroscience recording files

NWB stores neuroscience data and descriptions. In the paired source files here, SET holds recording descriptions and FDT holds binary samples.

A stale unit in the NWB muscle source must not override verified processing. SET headers match source hashes; FDT previews have checked local byte ranges rather than full verification of each upstream file.

Current registry lookup labels: Neurodata Without Borders file; NWB; Recording header and sample file; SET and FDT files; SET; FDT; EEGLAB header; byte-range preview

NeuroTransfer

Specify what would move between which systems, then test what the recipient can do with and without assistance.

The proposed transfer of specified task-relevant information or a mapping between declared systems, with assistance and retained learning evaluated separately.

The label does not establish a successful biological transfer or retained skill. A weight copy, recorded replay, guided movement and independently retained learning answer different questions. Each proposed interface needs its own source, target, task, controls, rights and qualified outcome.

Current registry lookup labels: neurotransfer

NeuroTranslate

Brain2Body Neural Activity is the long-form component label paired with NeuroTranslate in this proposed architecture. Recorded activity is an observation. NeuroTranslate names the mapping operation that could relate specified observations to movement, or reconstruct specified recorded features in the reverse direction. The observation, learned relationship and predicted value remain different things.

State the observations, the target and the task, then test how well the mapping works.

The proposed operation of mapping specified neural and body observations between declared inputs and targets in a stated task.

A model could predict movement from specified observations, or reconstruct specified recorded features in the reverse direction. Each direction needs its own test in the stated task. Neither establishes a unique hidden state, sensory delivery or biological skill transfer. NeuroTranslate names this proposed mapping operation; NeuroTranslation names the field and NeuroTranslator the implemented software.

Current registry lookup labels: neurotranslate; Brain2Body Neural Activity

NeuroTranslation

How could a particular neural recording be related to a defined movement task? NeuroTranslation is the proposed study of mapping nerve signals and the patterns that describe them between clearly defined situations like these. Every mapping must state where its inputs come from, what its outputs mean, and how it is tested.

A proposed field for translating or mapping neural signals and representations between explicitly defined source and target domains, with stated context, scale, tests and limitations.

Each connection must be assessed against its named source and stated test. The project has not demonstrated a universal brain language or an uploaded skill.

Scope: macro: Relationships between neural systems, body systems and defined tasks or organisms; micro: Mappings involving identified local signals, channels, features and circuit-level representations.

Organizational level: whole system; network; organ; local circuit; signal/feature

Current registry lookup labels: neurotranslation; neural translation; Connecting neural recordings to defined tasks; Neurotranslation

NeuroTranslation Network

A plan for connecting more neural and body components while keeping the origin and meaning of each value clear.

The proposed network would connect neural and body components while carrying the origin and meaning of each value through the mapping. Each link could then be examined in its own recording context. Body-wide translation and biological teaching remain unproved.

Current registry lookup labels: Proposed larger design; Source-Conditioned Neurotranslation Architecture; SCNA; Neurotranslation Network

NeuroTranslator

The software that joins particular recordings, prediction models, playback and computer-body checks.

Predictions and playback can be checked against the particular recordings they use. That is the scope of the implemented NeuroTranslator; neither a universal brain language nor skill upload has been demonstrated.

Current registry lookup labels: Working translation software; Neurotranslation Engine; NTE; Neurotranslator

NeuroTwin Trainer

A biological trainer would guide a trainee during a stated task.

A proposed biological trainer role in a declared Brain2Brain NeuroTransfer study.

The requested name alone establishes neither genetic twins, a validated digital twin, communication nor retained learning. Biological electrical activity is electrochemical and can include graded voltages and spikes; identify any actual measurement, otherwise keep the activity illustrative. NeuroTwin Tutor is a historical label; NeuroTwin Trainer is the current name. These names belong to a proposed programme view. They establish neither an implemented interface nor a demonstrated learning result.

Organizational level: whole system; signal/feature

Current registry lookup labels: NeuroTwin Tutor

Numerical arrays, saved arrays and prediction

NumPy provides tools for numerical arrays. NPZ saves arrays; inference means running a trained model to produce an output.

Portable execution still needs the correct processing and source identity. Matching saved-model outputs does not establish success on new organisms or biological skills.

Current registry lookup labels: NumPy inference and NPZ archive; NumPy; NPZ; portable inference; inference

Numerical checks on allowed movement

Checks that the computer model stays defined and within its stated joint limits.

Joint-limit penetration means going past an angular bound, not entering tissue. The 0.001 rad tolerance is a numerical rule, not a certificate of correct neural commands.

Current registry lookup labels: Mechanical feasibility and anatomical tolerance; anatomical limit; limit penetration; finite state; joint range; numerical tolerance

Numerical software and its data file

MATLAB is numerical-computing software, and MAT is a format for its structured data.

The second Indy recording uses the original MAT source. The format does not prove matching cells, past-only source processing or complete observation.

Current registry lookup labels: MATLAB and MAT file; MATLAB; MAT

Observe it, connect it, keep it useful

Observation burden, prediction on excluded cases and lasting task benefit need separate measurements. A clear picture does not answer those experimental questions.

Current registry lookup labels: Access it, Ground it, Keep it; access; grounding; stability; signal stability

One multiplier for modeled lengths

A single number chosen to reduce differences between the model’s muscle lengths and the source-derived length references.

The number is fitted on training poses and checked on distinct observed poses. It is not complete body calibration; no movement-control experiment is run on the scaled candidate.

Current registry lookup labels: Scalar length fit; global length factor; uniform factor; a*; scalar bridge; recording length-fit candidate; SDK-scaled candidate

One recording period

One experiment or recording period with its own subject, channels, timing and task.

Two recording days, or recordings from different animals, do not form one observed nervous system simply because the files are loaded together.

Current registry lookup labels: Recording and session; source recording; session; recording family

Open body-model collection

MyoSim supplies computer-body models with muscles and bones, including models used in this project.

The collection is earlier work, not a new project invention. Availability of an arm model does not establish matching macaque and human muscles or neural commands.

Current registry lookup labels: MyoSim, MyoSuite and MyoArm; MyoSim

Open muscle-control task tools

MyoSuite provides tasks and tools for experiments with muscle-driven computer-body control.

It is earlier software reused here. Success on a simulated task is not proof of biological learning or a recovered neural command.

Current registry lookup labels: MyoSuite

Other published models of neural recordings

NDT2 learns patterns across recordings made inside the cortex, the brain’s outer layer. POYO handles differing sets of recorded neural features by representing events and learning relationships between them.

The paper cites these as earlier work. It does not claim to have implemented them or use their existence as proof of universal transfer or matching cells.

Current registry lookup labels: NDT2; large-scale neural pretraining; Published model for varying recorded populations; POYO; neural-event tokenization; cross-attention bottleneck

Passing information and gaining a lasting skill

Receiving useful task information differs from acquiring a durable ability that works independently in new conditions.

A short software label budget is not measured biological practice time. Sewing, cube solving and breakdancing remain future examples without matched skill recordings here.

Current registry lookup labels: Task information transfer and skill acquisition; task transfer; skill transfer; skill upload; compressed acquisition; learning speed; learning; physical skill learning; skill acquisition

Past-only inputs and processing that uses later data

The local model uses completed past inputs. Some earlier source processing may still have used later samples.

Past-only indexing cannot reverse earlier filtering or rescaling that used the whole recording. It does not certify the full measurement-to-output delay in a living system.

Current registry lookup labels: Causal indexing and acausal preprocessing; causal; algorithmic support; acausal filtering; offline interpolation; upstream filtering delay

Peripheral nerve electrical recording

A method for recording electrical signals associated with peripheral nerves, called ENG.

The human preview contains aggregate electrical observations. It does not provide separately identified nerve-cell spike trains or a complete voluntary command sequence.

Current registry lookup labels: Electroneurography; ENG; surface ENG; peripheral recording

Phenotype

The trait is what must be observed: for example, a specified body characteristic or response in the setting being studied. Phenotype names these observable traits, rather than the genetic description alone.

The observable traits of an organism arising through genotype, environment and development.

State the actual trait, measurement method, developmental and environmental context. A phenotype is not a universal function that can be read uniquely from structure or sequence. Retain the distinction between a measured trait and a model's estimate.

Current registry lookup labels: No additional label declared.

Placing body parts from joint positions

Calculating where body parts and muscle routes lie when joint positions are given.

The formal name is forward kinematics, or FK. It places geometry; it does not calculate muscle-driven motion through time or recover muscle commands.

Current registry lookup labels: Forward kinematics; FK; kinematic replay

Planned commands, corrections and constant input

A planned command is feedforward; adjusting it from current model information is feedback; a constant muscle input is tonic control.

The human correction can change the bounded command goal. Recorded movement stays a scoring reference, and the native correction candidates remain rejected against constant-input comparisons.

Current registry lookup labels: Feedforward, feedback and tonic control; feedforward; feedback; corrected controller; tonic excitation; constant tonic

Prediction score compared with a mean-value guess

R^2 compares squared prediction errors with a constant mean-value reference. Higher values usually indicate better prediction on the stated data.

The score can be negative. Different recordings, target processing and ways of combining outputs prevent treating all R^2 values as directly comparable.

Current registry lookup labels: Coefficient of determination; R^2; R2; R^2; variance explained

Processing within spinal and peripheral pathways

Body-related information can be processed locally in spinal and peripheral structures rather than passing through one simple unchanged wire.

A cortex-only prediction does not observe those local processes. The separate recordings and computer-body checks do not establish a complete measured brain-to-body causal chain.

Current registry lookup labels: local spinal processing; local peripheral processing; spinal/peripheral processing

Programmable neurobiology

The proposed ability to specify a bounded nervous-system or sensorimotor outcome, select a qualified intervention, predict its effects, measure what happens and adapt within a validated closed loop.

Scope: Brain, spinal, peripheral, muscle, mechanical and sensory feedback roles need separate evidence. Bidirectional prediction is different from safe sensory/action writing. An intermediary may improve a mapping without making it one-to-one. Arbitrary skill upload or universal disease/disability repair is not demonstrated.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Current registry lookup labels: No additional label declared.

Programmable synthetic biology

Engineering specified biological behaviours by combining design, context-dependent molecular prediction, experimental measurement and iterative correction.

DNA/regulation, RNA/cell state and protein/interactions are explanatory layers, not a sufficient literal three-model recipe. Metabolism, feedback, environment and measured function matter. AlphaGenome and AlphaFold do not jointly certify whole-cell programmability; AlphaTranscriptome is a desired capability placeholder here.

Organizational level: Declared system or interface

Computational granularity: Declared process/interface/model

Statistical aggregation: Validation domain stated independently of physical scale

Current registry lookup labels: Programmable molecular biology

Public reaching recordings

Public files pairing cortical electrical events with reaching behavior. The sources use names such as RTT and MC_RTT.

The original model's release and the separately acquired Indy day have different processing. Their dates and target definitions must remain explicit.

Current registry lookup labels: Reaching source archive; RTT; MC_RTT; MC_RTT; O'Doherty reaching archive; Sabes reaching corpus

Public source records and document links

Repositories such as DANDI and OpenNeuro identify source versions. PDF is a document format, and a URL gives a resource location.

Source licenses and credit must be retained. A local paper draft or public input dataset does not mean the project's new code is published, licensed or submitted to arXiv or a journal.

Current registry lookup labels: Public records, document files and links; DANDI; OpenNeuro; PDF; URL; arXiv; public deposition; license; attribution

Published models for brain scans and scalp recordings

BrainLM learns from fMRI brain-scan data. Neuro-GPT combines an EEG input model with a model of changes over time and is discussed for motor imagery.

These are earlier methods for different measurements. The project does not claim to have trained them, make scans and electrical spikes interchangeable, or upload skills.

Current registry lookup labels: BrainLM; Published model of scalp signals; Neuro-GPT; NeuroGPT

Published neural-model comparison tasks

The Neural Latents Benchmark, or NLB, provides shared tasks for comparing models of neural recordings.

It informs the choice of comparisons. The paper's own local scores are not official NLB benchmark scores.

Current registry lookup labels: Neural Latents Benchmark; NLB

Published tests of changing recordings

FALCON is a published benchmark concerned with neural decoding when recordings change and recalibration is limited.

The paper uses related source recordings and its own data splits. Its local muscle-signal scores are not official FALCON scores.

Current registry lookup labels: FALCON benchmark; FALCON; FALCON M1; FALCON M1-A

Ranges that need statistical assumptions

A confidence interval gives a range for a quantity being estimated. A conformal prediction range tries to contain a future actual result at a stated frequency when its required assumptions hold.

The transfer figure shows one fixed run. The paper's ranges are based on earlier prediction errors, with related errors across time; they do not guarantee how often future true values will fall inside.

Current registry lookup labels: Confidence interval and conformal coverage; confidence interval; repeated-seed interval; Prediction range with tested coverage; conformal; conformal coverage; prospective-conformal; distribution-free coverage

Recent-signal model

A model component that keeps a learned memory of recent inputs as it predicts the next result.

Its formal name is gated recurrent unit, or GRU. This established method uses completed input history; its memory does not establish a universal neural language.

Current registry lookup labels: Gated recurrent unit; GRU; compact recurrent model; recurrent network

Record of what was observed

A record showing which channels or intervals were actually measured and usable.

Missing observations stay missing; they do not become zeros. The second Indy source does not supply separate observation intervals for every recorded unit.

Current registry lookup labels: Observation mask; observation interval; invalid support; missing observation

Recorded neural events

A spike is an electrical event identified in a recording. A spike train is the sequence of those events over time.

A recorded event count is a whole number tied to one channel and time interval. A model's average expected count can be a fraction and is a different kind of value.

Current registry lookup labels: Spike and spike train; neural event; spike count; spike train

Records linking files, checks and claims

A manifest lists specified files and identities; a receipt records checks; an evidence matrix links a claim to its supporting record.

These records also keep failures and withdrawn results visible. A catalog listing does not show that every dataset was acquired or used to train a model.

Current registry lookup labels: Manifest, receipt and evidence matrix; manifest; source manifest; audit receipt; processing receipt; evidence matrix; withdrawal record

Relay

Pass focused work between agents, then check their results before accepting it.

A manual coordination protocol that passes bounded work and verified evidence between specialist sessions, with one integration owner and an independent acceptance reviewer. Each session follows the currently authorized model, effort and speed routing.

A specialist can work on a bounded question while an independent reviewer checks its evidence before the integration owner accepts it. This is a project coordination workflow, distinct from an established external standard or native Ultra mode. Efficiency and quality claims apply only to tested tasks; universal performance equivalence and a 50 percent weekly saving remain undemonstrated.

Computational granularity: finite delegated tasks and evidence artifacts

Current registry lookup labels: relay; manual Relay protocol; Relay coordination

Rescaled muscle signal

A processed summary of muscle electrical activity put onto a relative scale.

The 16 targets have no physical voltage unit. A stale millivolt label in the source does not make them measured voltage, force or muscle activation.

Current registry lookup labels: Normalized EMG envelope; EMG envelope; quantile normalization; dimensionless EMG

Rescaling using training data only

Learning a rescaling rule from the training portion, then applying the same rule to later values.

This avoids using local test values to fit that rule. It cannot undo source processing that already used future samples or the whole recording.

Current registry lookup labels: Training-only normalization; scaler; target statistics; normalization

Response after an electrical stimulus

An electrical recording made while observing the response to a stimulus.

The human preview is a stimulus-response experiment, not voluntary movement. The stimulus itself can also leave an unwanted signal in the recording; that is an artifact.

Current registry lookup labels: Evoked potential and stimulus artifact; evoked stimulation; evoked recording

Results already seen during development

Results that have been inspected and can therefore influence later decisions.

The human 90% result and local motor test windows have this limitation. They can be reported, but their later reuse is not a new independent test.

Current registry lookup labels: Development evidence; development holdout; pilot; descriptive local test; inspected test

Reusing learning in another recording

Trying to use information learned in one setting to improve performance in another.

The comparison must state the new-setting labels and the controls. The reported frozen motor transfer and unadjusted human cross-day result do not establish successful biological or cross-species transfer.

Current registry lookup labels: Cross-recording transfer; transfer; cross-day transfer; cross-species transfer

Routing a muscle around a modeled obstacle

A rule bends a modeled muscle route around an obstacle. One obstacle shape, a torus, is shaped like a doughnut.

The correction restricts three elbow wrapping rules to a stated attachment segment. It removes reproduced route errors without deleting a muscle or claiming a new anatomical measurement.

Current registry lookup labels: Muscle wrapping and torus; wrap; wrapping route; torus; ELBOW_ULNA; ELBOW_ULNA; off-torus contact

Saved model and matching-output check

A checkpoint saves a model's state. A parity check asks whether another execution produces the specified matching output.

Repeating a result needs the same support, ordering, conversions, data splits, settings and body-model version. Passing software checks does not certify biological validity.

Current registry lookup labels: Checkpoint and parity; checkpoint; replay/checkpoint parity; model parity; reproduction

Scalp electrical recording

Electrodes on the scalp record electrical potential; the method is called EEG.

This is an aggregate electrical signal, not a list of individual cortical spikes. The project's paired preview follows electrical stimulation rather than voluntary motor training.

Current registry lookup labels: Electroencephalography; EEG; scalp potentials

Settings needed to repeat a run

The numbers and choices fixing one training run's starting point and procedure.

One fixed run can be reproduced. It does not show how results vary across repeated starts, animals or future recording days.

Current registry lookup labels: Random seed and configuration; seed; single-seed; fixed protocol

Shared internal model

A model can learn an internal pattern and reuse that pattern when making several predictions.

The paper calls the pattern a latent representation and the reusable model a backbone or trunk. The tested shared model and frozen-transfer candidates were rejected.

Current registry lookup labels: Latent representation and backbone; latent model; shared latent; backbone; trunk; shared parameters

Simple weighted prediction model

A linear model that combines inputs using fitted weights while discouraging excessively large weights.

The formal method is ridge regression. It provides a meaningful comparison: a more complex model must add value rather than merely produce a plausible display.

Current registry lookup labels: Ridge regression; ridge; linear control; linear model

Simplicity principle

Introduce one concept or element at a time, using separate slides or successive reveals according to what makes the relationship easiest to understand.

Introduce each new element on its own, then show how the elements connect. Use separate slides or successive reveals to make the logical relationship visible. Simplicity cannot remove scientific qualifications, evidence, source rights or required methods. Actual comprehension and exported readability still need review.

Current registry lookup labels: No additional label declared.

Skin-prick suit

A suit shows proposed interface contacts across the body.

The proposed body-side (peripheral) interface, drawn as a suit.

A contextual display label in the user's Interoceptomics visual instructions, not a general clinical synonym. The name establishes no acquired neural data, registration, tissue access, function, safety, clinical use, species or physical dimensions.

Current registry lookup labels: skin prick suit; peripheral suit; peripheral prosthesis; peripheral interface

Small connector model

A smaller fitted mapping that connects one recording to an existing model or interface.

The connector can change while the older model stays fixed. That does not establish matching neurons or useful transfer without new-setting calibration.

Current registry lookup labels: Adapter; linear adapter; recording adapter

Software that calculates a computer body's motion

MuJoCo calculates how a modeled body moves under forces and control inputs.

It is established software reused here. Its availability does not prove that the chosen body model or mapping matches a particular animal's physiology.

Current registry lookup labels: MuJoCo; dynamics engine

Spinal strip

A strip shows the proposed interface along the modeled spine.

The proposed spinal interface, drawn as a strip along the modeled spine.

A contextual display label in the user's Interoceptomics visual instructions, not a general clinical synonym. The name establishes no acquired neural data, registration, tissue access, function, safety, clinical use, species or physical dimensions. It does not validate a modeled dorsal-surface or tailbud overlay as biological spinal-cord or conus anatomy.

Current registry lookup labels: spinal strip; spinal strap; spinal prosthesis; spinal interface

Spine2Body

A muscle signal becomes a movement through tendons, joints, contact and load. Measure those steps together, and a failed reach becomes a specific question about activation, coordination or mechanics.

The pathway through which spinal output reaches peripheral motor nerves and muscles, producing action through the body's mechanics, while sensory signals return from the body.

Surface EMG measures muscle electrical activity, not peripheral nerve firing or force. Spinal plasticity, momentary reflex modulation and peripheral adaptation remain distinct. Simulated movement is a prediction to compare with independent measurements.

Organizational level: whole system; network; signal/feature

Current registry lookup labels: spine2body; spine to body; spine-to-body

Stopping training under a stated rule

Ending training when the chosen development rule says to stop, rather than always using every allowed round.

Compared methods have equal round limits but can perform different numbers of actual updates. This is not measured biological learning time.

Current registry lookup labels: Early stopping; epoch cap; training update

Structured task text and supported instructions

JSON stores structured text values. The task compiler converts a supported instruction into a defined program.

The compiler chooses available recorded building blocks and limited computer-controller practice. It does not program arbitrary biological skills or supply missing task evidence.

Current registry lookup labels: JSON and task compiler; JSON; task JSON; natural-language compiler; supported primitive

Surface electrical recording over the spine

Surface electrodes placed over spinal regions record electrical potentials; the method is called ESG.

These measurements combine electrical contributions. They are not separately identified spinal-neuron spike trains.

Current registry lookup labels: Electrospinography; ESG; surface spinal recording; surface ESG

System 3 thinking

System 3 thinking is the partnership; a large movement model is the movement-focused model we aim to build within it.

The programme's human–AI partnership framing in which an external supervisory model helps interpret, train or calibrate task-specific neural/body mappings; the movement-focused model being built is called a large movement model.

Expert footage provides movement examples, but it does not supply the depicted experts' neural or muscle recordings. An external supervisor would help interpret, train or calibrate task-specific neural/body mappings while the large movement model remains the movement-focused model being built within that partnership. Shaw and

Nave use System3 for artificial cognition outside the brain in a different research account. This programme's task-specific framing establishes no new anatomical brain layer, biological neo-neocortex, third anatomical thinking system or achieved universal nervous-system model.

Organizational level: Declared game/model/interface role

Computational granularity: Declared task-specific model or interface

Statistical aggregation: Validation scope declared per experiment; no physical-scale inference

Source: Thinking—Fast, Slow, and Artificial

Source identifier: shaw-nave2026

DOI: 10.2139/ssrn.6097646

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6097646

Prior proposed System3 usage for artificial cognition outside the brain; not an established third anatomical brain system

Source: Thinking Fast, Slow, Artificially: AI and Your Brain

Source identifier: wharton-authors2026

https://executiveeducation.wharton.upenn.edu/thought-leadership/wharton-at-work/2026/05/thinking-fast-slow-and-artificially/

Primary institutional account of Shaw and Nave System3 research

Current registry lookup labels: System three thinking; System3 thinking

Testing on an excluded day, animal or species

Checking a model on complete identities that were kept out of the relevant development steps.

The paper uses formal names such as leave-session-out or leave-species-out. A good score within one session does not answer these broader tests.

Current registry lookup labels: Leave-out generalization; leave-session-out; leave-animal-out; leave-species-out; unseen-organism support

Testing possible next actions in the model

Simulating near-future actions before choosing the next command.

The copied test states must not change the live model. The native result remains rejected and too slow for real-time use; cached playback is a separate function.

Current registry lookup labels: Short-horizon planning and lookahead; horizon controller; lookahead; copied-state rollout; model predictive planning

The comparison needed to judge improvement

A defined comparison that shows whether the proposed method adds useful information or performance.

A simple predictor, random fixed model and constant muscle command answer different questions. Producing a plausible animation is not a substitute for beating the relevant comparison.

Current registry lookup labels: Baseline and control; simple baseline; tonic baseline; random frozen control; behavior-only predictor

Thought2Action

Begin with one intention and one action: reach for the cup. Measure the neural, muscular and mechanical steps between them. The aim is to make that intent-to-action relationship as close to one-to-one as possible within the declared task, person, state, context and tolerated error.

A task-conditioned mapping from a participant's stated or inferred intent to a specified action, evaluated through independently measured outcomes.

Intent inferred from EEG is not direct access to private thought. Movement constrains competing explanations; it does not certify an achieved universal bijection or the best possible mapping.

Organizational level: whole system; network; signal/feature

Current registry lookup labels: thought2action; thought to action; thought-to-action

Thought2World

Follow the intention all the way to its consequence, then follow sensation and perception back to the next intention. Measuring the middle makes errors easier to locate and the next experiment easier to define.

The composed pathway Thought2Action followed by Action2World: intent is translated into an action, and that action produces an outcome that can be independently checked.

This is a proposed programme pathway, not a claim of unrestricted thought control. It is not silently equivalent to the existing Body2World lookup label, Body2Action or Brain2Motion. External touch and movement tasks alone do not solve internal-body sensing or the full Interoceptomics programme.

Organizational level: whole system; network; signal/feature

Current registry lookup labels: thought2world; thought to world; thought-to-world

Timing relationships between spikes

How closely spikes can repeat and how spikes from different recorded features occur together.

The project checks predicted average counts. It does not establish these finer timing relationships or a complete account of biological spike noise.

Current registry lookup labels: Refractoriness and spike synchrony; refractoriness; joint spike synchrony; biological count-noise model

Touch brain area

A brain area that processes body-related sensory information. The source study records from somatosensory cortex, including Area 2.

The formal shorthand is S1. The selected model estimates an earlier limb state; it does not recover voluntary motor commands or spinal and nerve recordings.

Current registry lookup labels: Somatosensory cortex and Area 2; S1; Area 2; Area2_Bump; somatosensory

Trainer and trainee

The trainer supplies guidance; the trainee's later control is tested. An experienced trainer helping a novice asks a different question from two novices practicing together.

The proposed comparisons include experienced-to-novice, novice-to-novice, individual practice and matched no-link conditions. Later trainee-alone testing removes trainer commands while retaining any bridge required to express movement.

Current registry lookup labels: Tutor and pupil; biological tutor; artificial tutor; live tutoring; tutor withdrawal; experienced-to-novice tutor; novice-to-novice; individual practice; Trainer and learner; biological trainer; artificial trainer; live training; trainer withdrawal; experienced-to-novice trainer; Guide and learner; Guide and trainee

Training a fresh model

Training a new model for the current recording without importing an older model's learned main weights.

This is the scratch specialist. It performs better than the frozen-transfer candidate in both reported label budgets.

Current registry lookup labels: Scratch training; scratch specialist; new-recording scratch specialist

Training with neural samples without movement labels

Using neural samples for a training objective that does not require paired movement labels.

These samples still provide information and require computation. The first 5% label condition permits full training-neural support, so it is not only 28.5 seconds of total observation.

Current registry lookup labels: Neural-only training support; neural-only pretraining; unlabeled neural support

Trying to move and actually moving

Trying or being cued to move is different from measuring the movement that occurred.

The human labels describe attempted finger movement. They do not independently separate cue from intent or provide executed finger trajectories.

Current registry lookup labels: Attempted versus executed movement; attempted finger identity; cued attempted movement; executed kinematics

Typical error on the original scale

RMS or RMSE squares the deviations, averages them, and takes a square root so the result has the original unit.

An error per joint differs from a combined seven-joint error. An error against a changed command goal also differs from an error against the same recorded reference.

Current registry lookup labels: Root mean square error; RMSE; RMS; root-mean-square; tracking error

Unassigned event channel

A channel collecting events not assigned to the sorted features used by the model.

These events are excluded from the 171 sorted features in the second Indy recording. They cannot silently become another identified cell.

Current registry lookup labels: Residual or hash cell; residual/hash; hash slot; threshold crossing

Units for distance, speed and angle

Metres, centimetres, millimetres and micrometres measure distance. Metres per second measures speed; degrees and radians measure angle. Radians per second (rad/s) measures the rate of change of angle.

One metre is 100 cm or 1000 mm; one mm is 1000 µm. Right-angle and varied-angle maze layouts differ in geometry, not in a validated difficulty multiplier. Unit conversion or a small model error does not prove anatomical or decoded-pose accuracy.

Current registry lookup labels: Length, velocity and angle units; m; metre; cm; centimetre; mm; millimetre; µm; micrometre; um; m/s; rad; radian; degrees; °; rad/s; orthogonal; nonorthogonal; varied-angle maze; maze layout

Units for pushes, turns and electrical potential

Newtons measure force; newton metres measure turning moment; millivolts measure electrical potential. A dimensionless value has no physical unit.

The handle record mixes force and moment units. The rescaled muscle targets are dimensionless despite a stale mV label and cannot be read as force or activation.

Current registry lookup labels: Force, moment and voltage units; N; newton; N m; N·m; newton metre; mV; millivolt; dimensionless

Units for time and sample rate

Seconds measure time; milliseconds are thousandths of a second. Hertz counts repetitions or samples per second.

A 250 Hz source sample interval is 4 ms; a 500 Hz simulation interval is 2 ms. These rates differ from playback pace, biological travel time and time spent computing.

Current registry lookup labels: Time and frequency units; s; second; seconds; ms; millisecond; milliseconds; Hz; hertz; sample rate; 250 Hz; 500 Hz

Unwanted stimulus signal in a recording

Part of a recording caused by the stimulus or measuring setup rather than the intended response alone.

Stimulus artifacts remain in the human previews. Their presence and the evoked task limit interpretation as voluntary motor evidence.

Current registry lookup labels: stimulus artifact; artifact

Use in a new setting without labeled adjustment

Applying a learned model to a new setting without using its labels to adjust the model.

The second M1 model uses explicit new-recording labels. Its test score is a result after calibration, not zero-shot success.

Current registry lookup labels: Zero-shot transfer; zero-shot

Velocity estimated from earlier positions

Two recorded fingertip positions can give an estimate of how fast the fingertip moved and in which direction between the samples. The signed components describe velocity; speed is its non-negative magnitude.

A position-derived velocity estimate formed from the difference between current and earlier recorded positions divided by their elapsed time; individual components retain sign. Speed is the magnitude of the velocity vector.

In the separate calibrated motor recording, each interval uses the last available fingertip sample and the actual elapsed time to form a velocity target in m/s. This makes the calculation causal with respect to available position samples, not proof that the tracker supplied them without filtering or delay. Upstream filtering and latency remain unestablished; the derived target is not an original, directly supplied velocity trace.

Current registry lookup labels: Backward finite difference; backward difference; finite difference; causally derived velocity; 250 Hz position support; Speed estimated from earlier positions

Velocity of each model movement

The model’s stored velocity for its movement degrees of freedom, including direction.

The formal variable name is qvel. Its signed generalized velocities depend on the model's degrees of freedom; speed is a magnitude. Component units and frames depend on the declared coordinate and model. In MuJoCo, quaternion orientations can make qpos and qvel different lengths, so qvel is not an entry-by-entry derivative of qpos.

Current registry lookup labels: qvel; generalized velocity; joint speed; Speed of each model movement

What a model estimated

A model's estimate of a defined result, such as speed, from the inputs it was given.

An estimate remains an estimate even when it matches the test data well. It does not become a new measurement of the spinal cord, nerves or muscles.

Current registry lookup labels: Decoded or inferred prediction; decoded; inferred; reconstructed; prediction

What an instrument recorded

A value produced by a stated measuring method, such as an electrode or a movement tracker.

Recorded data can contain noise, filtering or artifacts. Calculating muscle length from tracked markers does not turn that length into a direct tissue measurement.

Current registry lookup labels: Recorded measurement; measured; observed; recorded signal

What the computer body produced

A value produced while running a computer model, such as a modeled muscle's activity.

Computer-body movement and forces can be checked for numerical errors. They remain separate from measurements of a living body and from evidence that an organism learned.

Current registry lookup labels: Simulated quantity; simulated; numerical model state

What was calculated from tracking

A value calculated from measurements together with a body model, such as an estimated joint angle.

The study's joint angles and muscle lengths use tracked movement and a model. They are not direct recordings of muscle commands, tissue length or peripheral nerve spikes.

Current registry lookup labels: Model-derived quantity; tracking-derived; source-derived; biomechanical covariate

Where a value came from

The record of a value's source and the steps used to produce it.

For example, a measured hand position, an angle calculated from that position and a computer-generated muscle state have different origins. The software must preserve those differences.

Current registry lookup labels: Provenance; lineage; source integrity

Which recorded channel is which

The labels and order used to identify neural features within one recording.

The labels do not identify the same neurons across days. The project did not accept unverified matches between cells in different recordings.

Current registry lookup labels: Source-local unit identity; ordered channel identity; cell correspondence; unit order; neuron index

Which samples a value depends on

The input history and exact time interval needed to produce a value or target.

A single timestamp does not show all of this support. Overlapping histories and invalid samples near data splits need explicit checks.

Composed pathways

The proposed intermediary has a positive purpose: understand and measure the processes between intent and action so the task-specific mapping can be made as close to one-to-one as possible. The task, participant, state, context, target and tolerated error must be specified and independently tested. This is an aim for identifiability and control, not an achieved universal bijection or a demonstrated optimum.

Form → function → action and intent → action describe connected questions. The following proposed pathways separate the links that would need independent tests before their combined prediction could be trusted.

Biological pathway: Brain2Spine → Spine2Body; resulting in Brain2Body

Relate brain activity to spinal processing, and spinal/peripheral processing to the body; compose the tested links into an intent-to-body-action pathway.

Functional pathway: Thought2Action → Action2World; resulting in Thought2World

Relate a declared intention to a measured action, and that action to its measured consequence in the task environment.

Molecular analogy: DNA → RNA; RNA → protein; resulting in DNA → protein

The intermediate processes explain how an initial specification reaches its downstream product. This analogy illustrates the need for a measured middle; it does not assert identical biology or a context-free one-to-one relationship.

System 3 engineering route: Neural activity → Musculoskeletal simulation → Programmable model

The proposed engineering route to NeuroTranslation, coordinated by the System3 human–AI supervisory model.

Paired source-bound brain/spinal/peripheral/muscle and sensory/outcome observations constrain mechanics and a task-specific learner. A musculoskeletal simulation represents the biomechanical middle. Model read/write capabilities must be qualified individually.

Intent → action → outcome returns through sensory feedback to perception and subsequent intent; actual observed/inferred/simulated links remain distinguishable.

Interoceptomics combines interoception and connectomics. Touch, proprioception and external sensorimotor tasks can support this programme; they do not by themselves demonstrate internal-body sensing. Combining the proposed pathways advances a research aim rather than establishing a comprehensive solution.

Brain2Body names the wider programme. Body2World remains a contextual lookup label for Body2Action; Brain2Motion and Body2Action retain their own scopes. These related names do not make their measurement targets equivalent to those of the five composed paths.

The physical spinal pathway remains read-only in this proposal. Each claimed recording or writing capability needs its own independent qualification.

Current registry lookup labels: Temporal support and context; support; history window; context; next-target support; overlapping histories

Whole Body Interface

The hand is one part of the question. Whole Body Interface names the wider ambition to connect declared body systems, while showing which pathways have actually been observed and which remain missing.

A proposed extension of brain–body interfacing toward specified observations and outputs across the brain, spinal, peripheral and body systems.

Whole body states an ambition, not achieved coverage. An interface exchanges specified observations or outputs; the existing model spanning the body is a related

modelling ambition, not the same object. State the recorded systems, missing pathways, units, clocks, frames and output routes. Separate datasets can support qualified learning and transfer, but they do not become simultaneous measurements of one body. Physiological input/output direction, software input/output and ordinary sensory feedback remain distinct; no spinal writer or physical actuation is authorized.

Current registry lookup labels: No additional label declared.

XenoTransgenesis

Which genetic feature would be studied, in which organism, and what would count as the relevant change? XenoTransgenesis is the user's working label for that proposed question within EndoHumanisation.

A working roadmap name for proposed research involving specified genetic components from another species in a model organism.

XenoTransgenesis is the preferred spelling; XenoTransgenicism remains a historical lookup label. This is not an established replacement name for every form of transgenesis, genome editing or humanisation; use the established term appropriate to any actual manipulation. Neither a genetic change nor its name establishes the intended functional effect or human equivalence. No procedure, approved organism, developmental schedule or experimental permission is defined here.

Current registry lookup labels: XenoTransgenicism

A cross-species graft

A chimera has biological contributions from different sources. A graft is added tissue; when it crosses species, it is a xenotransplant. An organoid is an organized tissue model.

Survival, physical connection, activity and useful function are separate questions. The proposed comparisons do not establish a humanized whole brain or an implemented treatment.

Cross-species-graft component only; not a strict synonym for every chimera, graft or organoid, or achieved ExoHumanisation.

Additional lookup label: XenoTransplantation

Proposed names and explanatory comparisons

Neo-neocortex

A metaphor for an external machine contribution to human cognition; no anatomical addition is claimed.

AlphaTranscriptome

AlphaTranscriptome is a proposed name for a desired RNA/cell-state prediction capability. It is not an adopted general-purpose model. AlphaGenome already predicts some expression and splicing [1]; other RNA/cell-state models have bounded domains. These precedents leave whole-cell programmability to be independently demonstrated.

AlphaCell

A label for integrating cellular layers appears in the supplied passage; no implementation, novelty or product entitlement follows.

Connetype

A connection profile would describe the selected observations for one individual. Its anatomical coverage, resolution, method and uncertainty must remain visible. The profile does not identify that person's unique function, thoughts or a fixed biological type.

An individual's specified neural-connection profile under a declared measurement and representation.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

The tentative name is Connetype. Connectype remains an unapproved editorial alternative; it is neither a lookup alias nor an adopted spelling.

Interotype

A heartbeat-related observation can vary with state and measurement conditions. A proposed individual profile must state what was observed and how repeatability would be assessed; it cannot treat touch or proprioception as interchangeable with interoception.

An individual's profile of specified internal-body sensing observations under declared conditions.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

Interotype is a tentative name, rather than an established scientific type.

Enactotype

A profile of selected actions is bounded by the tasks, observations and conditions used to construct it. Repeatability, state dependence and generalization remain empirical questions; a stable innate action type or inferred unique intent is not established.

An individual's profile of specified actions and action–outcome observations in declared tasks.

The original three-name line ends in a question mark; all three remain tentative. No formal empirical definition or repeatability threshold is fixed.

Enactotype is a tentative name, rather than an established scientific type.

Form, Function, Feature

Start with a structure, ask what it does in the stated setting, then name the observable feature being compared. A hand's form, its grasping function and a measured property of a particular grasp answer related but different questions.

Feature must identify the actual observed or defined trait in its context; it is not a new universal third biological layer or a substitute for function or action. The existing Form → function → action account and intent–action distinction remain unchanged. Structure constrains function; context, history, selection and co-option prevent a guaranteed one-to-one or teleological chain.

Semiosis as a sensorimotor analogy

A drawing can resemble a hand, a pointing finger can indicate an object, and an instruction can follow a learned convention. Such signs can guide action through different relations; they do not assign perception, sensation and action to three exclusive boxes.

Icon, index and symbol describe sign–object relations, while sensation, perception and action name processes. A sign can combine relations. No one-to-one biological correspondence, neural code, private-experience identity or strict isomorphism is established. Outside-in and inside-out perspectives must state whether they concern physiology, modelling or interpretation.

Agent vs Neurology

The roadmap ends by asking what an agent could contribute to a defined neurological problem. Agent vs Neurology retains that proposed destination; the specific problem, task and comparison still have to be chosen.

An unresolved Stage 5 research question, not a selected disorder, intervention, diagnostic system or clinical trial.

No specific disease, patient population, endpoint, treatment mechanism, universal cure or successful clinical benefit is inferred. Any future study must define those

matters and its ordinary feedback or separately qualified physical interface under the original scientific, participant, rights, safety and independent gates. A prior software, metadata or model result cannot supply this evidence.

Agent vs Neurology names Stage 5 of the proposed roadmap.

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A Reference Comparison Protocol: Unfilled Record

A moving hand gives the instruments something in common to answer to. Scalp EEG would observe electrical activity at declared scalp sites; surface EMG would observe muscle electrical activity at declared arm sites. Neither tells us, by itself, where the hand went. The proposed LEFT and RIGHT tripod views would provide complementary movement observations. Each acquisition needs its own identifiable series of real references, followed by an independent comparison of the assembly. Two streams can agree because they share the same mistaken calibration.

This is an unfilled protocol and record. No observations, admitted reference series, measured errors or acceptance decisions have been entered. The apparatus and its anatomical and functional claims remain unvalidated. The older above/below maze illustration retains its separate schematic context, perpendicular-to-floor geometry and material-path questions. Its directions are not synonyms for the current LEFT/RIGHT tripod arrangement. Actual positions, viewing paths, coverage and transmission would need evidence.

Identify the body and the observation

A photograph can constrain a visible proportion; it cannot supply an EEG channel or a calibrated trajectory. Record each reference's provider, stable identifier, version or acquisition context, checksum if materialized, license, access terms and supporting claim. Identify the organism, developmental stage, participant or specimen, session, body part, task and view or time interval. Record physical dimensions and units, posture, support and contact, so that the comparison concerns the same body and movement. Keep those dimensions separate from computational detail and from the rule used to average errors. Existing datasets, atlases and photographs retain their separate study identities; none is silently admitted for this apparatus.

EEG needs sites, reference, channel order and electrical units. Surface EMG needs named muscles, sleeve and contact identities, reference and electrical units; it supplies neither incoming sensory-nerve traffic nor direct force. Each tripod view needs its selected modality and camera or ranging geometry. Optical triangulation and LiDAR ranging measure different things. Preserve raw observations, derived features and model estimates separately. These recordings do not identify hidden spinal activity, individual neurons or a unique brain state.

Make time and position comparable

A reach can follow the measured path and still arrive late. Record source timestamps, clock identities, cadence and units, synchronization events, offsets, drift, dropped samples and interpolation. Retain calibration observations, frame transformations, their uncertainty and checks against an independent reference. Report offline processing and end-to-end latency explicitly. An alignment error must not disappear inside a movement score, and no real-time capability follows from this protocol.

Compare scalp EEG, each arm's surface EMG, each tripod acquisition, and the time/frame calibration and any actual material path against their respective series of real references. Keep overlap, occlusion, dropout and failed views alongside anatomy, proportion, support and contact. Preserve both motion observations and the joined map. A missing finger remains missing unless its replacement is labelled as an estimate. The assembled map then needs its own independent movement comparison; agreement among its inputs cannot establish the accuracy of their common reconstruction.

Test what the assembly claims

Independently measured movement can test a virtual body's path. Contact and force claims need appropriate measurements of contact and force; a lifelike image cannot establish valid mechanics. A learner's prediction needs a withheld measured target, declared input availability, baselines, negative controls, independent held-out participants, sessions, tasks or species where applicable, and leakage checks. Keep the reserved recordings closed and the original tolerances unchanged. This record admits no new fitting, biological teaching, participant acquisition, stimulation, physical intervention or model promotion. Physical spinal scope remains read-only.

For each component and the assembled apparatus, retain matched reference views or time intervals, the target observable, result, error with units, repeatability, uncertainty and failures. Declare the applicable original acceptance rule, its source, domain, comparator and tolerances before evaluation. The programme's unchanged real-reference contract and every stronger original scientific, anatomical, functional, physical, held-out, rights, visual, runtime and export criterion still govern their respective claims. No universal numeric threshold is introduced or relaxed. Biological function and clinical readiness need evidence beyond a matched motion record.

Unfilled comparison record

Use one record per reference and comparison, retaining a series for every component and for the assembled map. All values below are `null`; reference-series lists are empty and every acceptance state is `HOLD`.

Source and rights. Record ID, component or assembly, provider, source identifier, version or acquisition context, local checksum if materialized, license, access and supporting claim.

Body and task. Organism, developmental stage, participant or specimen, session, body part, task, view or time interval, physical dimensions and units, posture, support, contact, actual sensor geometry and material path.

Acquisition. Modality, measured quantity and units, sites, montage, reference, channel and contact identities, source frame and clock, timestamps, cadence, synchronization events, offsets, drift, calibration observations and transforms, calibration uncertainty, interpolation, missingness, occlusion, offline processing and end-to-end latency.

Independent comparison. Component-series and assembly-series identifiers, independent reference and its acquisition, aligned interval, raw/derived/simulated provenance, input availability, held-out split and custodian, baseline, negative controls, leakage checks and target observable.

Result and decision. Result, error and units, repeatability, uncertainty, failure cases, applicable original acceptance rule and source, domain, comparator, tolerances, independent reviewer, evidence and verdict.

No measured series or scientific validation is supplied by this empty record. Rights, physical, clinical, runtime, native, render, export and publication acceptance remain separate and open; no hosted release is claimed.